

Drug Dosing Considerations in Patients With Acute Kidney Injury and Chronic Kidney Disease

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CHAPTER OUTLINE

EFFECTS OF AKI AND CKD ON DRUG DISPOSITION, 1990

PHARMACOGENOMICS, 1996

PHARMACODYNAMICS, 1997

ASSESSMENT OF KIDNEY FUNCTION, 1998

DRUG DOSING CONSIDERATIONS, 2001

CLINICAL BOTTOM LINE, 2007

KEY POINTS

- Chronic kidney disease (CKD) results in alterations in the absorption, volume of distribution (V_D), metabolism, and elimination of most drugs.
- The V_D of many drugs is increased in the presence of acute and CKD as a consequence of volume expansion and/or reduced protein binding.
- In addition to the expected decrement in renal clearance, nonrenal clearance (i.e., gastrointestinal and hepatic drug metabolism) of several drugs is also reduced in CKD patients.
- Individualization of a drug dosage regimen for a patient with reduced kidney function is based on the pharmacodynamic/pharmacokinetic characteristics of the drug, the patient's degree of residual renal function, and his or her overall clinical condition.
- Patient covariates, disease covariates, and pharmacogenetics all contribute to variability in drug disposition.
- The drug dosing guidelines for CKD patients in many drug information resources are highly variable and many are not optimal for clinical use.
- The effect of hemodialysis, peritoneal dialysis, and continuous renal replacement therapy on drug elimination is dependent on the characteristics of the drug and the dialysis prescription.
- Prospective monitoring of serum concentrations is often warranted, especially for narrow therapeutic index drugs.

It is estimated that 10% to 15% of the global population had chronic kidney disease (CKD) and the number of deaths from CKD has risen by more than 80% in the past 2 decades.^{1–3} The prevalence varies widely across the world, in part, because of differences in the prevalence of CKD; heterogeneity of the laboratory methods used to detect CKD; and environmental factors, public health policies, and genetics.³ The incidence of CKD has more than doubled in the past 20 years in adults older than 65 years.⁴ This is

due to age-related reductions in kidney function, multiple medical comorbidities, increases in the prevalence of older individuals who have survived cardiovascular disease, and an increased use of medications that alter kidney function, among others.⁵ CKD can affect multiple organ systems, and the associated pathophysiologic changes can profoundly alter the pharmacokinetics (PK) and pharmacodynamics (PD) of many drugs.^{5–7} Clinicians must assess kidney function and consider how it alters the disposition of drugs and their

active or toxic metabolites to optimize the health outcomes of their patients.

The epidemiology of acute kidney injury (AKI) varies widely due to the variability in the criteria used to evaluate the patient.⁸ A the disorder has a frequency of 2% in hospitalized patients, its prevalence among the critically ill is up to 30 times greater.^{9–13} The number of patients with AKI has increased in the past 10 years in most regions of the world.¹³ The development of some degree of CKD and/or need for chronic renal replacement therapy (RRT) are increasingly common among some survivors of AKI.¹⁴ The widespread use of renal replacement therapies for treating patients with AKI (e.g., continuous venovenous hemodiafiltration) and end-stage kidney disease [ESKD; hemodialysis (HD) or hemodiafiltration] during the past decade mandates an understanding of their influences on drug disposition in the high-risk patients.^{15,16} The advances in dialyzer technology and delivery systems have resulted in enhanced efficiency of dialysis treatments for many patients. Unfortunately, rarely have these advances been accompanied by reevaluations of the influence of dialysis on drug disposition.^{6,7,17} Although innovation in peritoneal dialysis has been more modest, few studies have examined the effects of newer adequacy targets, or the use of non-dextrose-containing peritoneal dialysates on drug disposition.

Medications that are predominantly eliminated unchanged (fractional extraction [FE]) by the kidney may accumulate in CKD patients, which can increase the risk of adverse effects. As a general concept, if 30% or more of a drug is eliminated unchanged in the urine, it will have a high likelihood of requiring dosage regimen adjustment in CKD patients, especially those with stage 3 to 5 disease.^{5,7,18} The PK of drugs with an FE less than 30% also may be affected and thus require a dose adjustment. In fact, 32.2% of such drugs approved in the United States from 1998 to 2010 had a dosage-adjustment recommendation for CKD patients in the product labeling.¹⁹ Despite an increasing number of renal studies by the pharmaceutical and biotechnology industry and improvements in approved product labeling language, challenges remain for dose adjustments in CKD patients, especially for oncology and antiretroviral agents.²⁰

Although PK and PD studies in CKD patients are done for an increasing percentage of drugs, for up to 70%+ in the United States in the past decade the results are often not published in a timely fashion, if at all.¹⁹ Regulatory authorities have not yet responded to the challenge of ensuring robust data for patients with impaired kidney function by requiring a PK and PD investigation plan as a routine part of drug development.⁶ Furthermore, patients with moderate to severe CKD are typically excluded from participation in major safety and efficacy studies required for drug registration. Therefore the usefulness of the dosage recommendations, if developed early, cannot be evaluated in those with the disease or condition for which they are intended. Data on the use of many drugs in patients with CKD, as well as the impact of dialysis, are often limited or absent for years after regulatory approval. If there is no official dosage regimen recommendation in the product labeling, an adjustment may be calculated on the basis of the drug's FE and the ratio of the patient's residual renal function relative to an age and gender normal value for estimated creatinine clearance (eCrCl) or estimated glomerular filtration rate (eGFR).^{21,22}

The pharmaceutical industry began to investigate the relationship of kidney function to the PK and PD of the drugs they had in development in the 1970s. Regulatory guidance or clinical consensus on when investigations should be conducted and with what degree of rigor, however, did not appear until the late 1990s and they continue to evolve.²³ Thus much of the data on the PK of drugs in patients with kidney disease were the result of clinician-initiated, postmarketing studies. Significant differences exist with respect to the means of assessment of kidney function and classification of the degree of impaired kidney function in those studies that have been conducted.¹⁸ Studies conducted to meet global regulatory expectations have resulted in the generation of conflicting results for some drugs, in part, because of the differences in the ethnic composition of the patient population, the study design (single dose vs. multiple dose), and the data analysis methods that were used. This has contributed to marked variability among common sources for renal drug dosage recommendations.^{24–27} The availability of robust and reliable information to guide prescribing for patients with kidney disease remains challenging. Despite the availability of this extensive collection of published evidence, dosing errors in CKD patients still occur at an alarming rate.^{20,28–30} The expanded use of electronic medical records has not resolved the need for clinician proactivity to optimize the use of medications in CKD patients; in these studies, up to 85% of the medications ordered had nephrotoxic potential and greater than 20% of the drugs ordered were not dose adjusted for the patient's kidney function.^{28,31–33} Thus the optimization of therapy for AKI and CKD patients is ultimately dependent on the clinicians' utilization of the data that are available and in many cases presented to them via electronic medical record system alerts at the time of prescribing.

In this chapter, the influence of AKI and CKD on drug PK properties is characterized, and a guide for individualizing drug therapy in patients with AKI and CKD is presented. We also present dosage recommendations for many commonly used drugs. The role of PK measures alone, their use in combination with PD, and pharmacogenetic testing in drug dosage regimen design are all discussed. The impact on drug disposition on either maintenance dialysis for ESKD or continuous RRT (CRRT) for patients with AKI is discussed. Again, dosage recommendations for many critical drugs are presented.

EFFECTS OF AKI AND CKD ON DRUG DISPOSITION

PK describes the time course of drug absorption, distribution, metabolism, and elimination. PD provides a characterization of the complex interaction of drug concentrations, receptor–drug interactions, mechanism of action, and clinical factors, such as concurrent diseases and degree of organ dysfunction on patients' response to drug therapy. The combination of PK and PD drug characteristics allows clinicians with foundational information to make rational prescribing decisions.

When given intravenously (IV), a rapid decrease in the plasma concentration follows an initial high drug concentration. This decrease occurs as the drug distributes from the

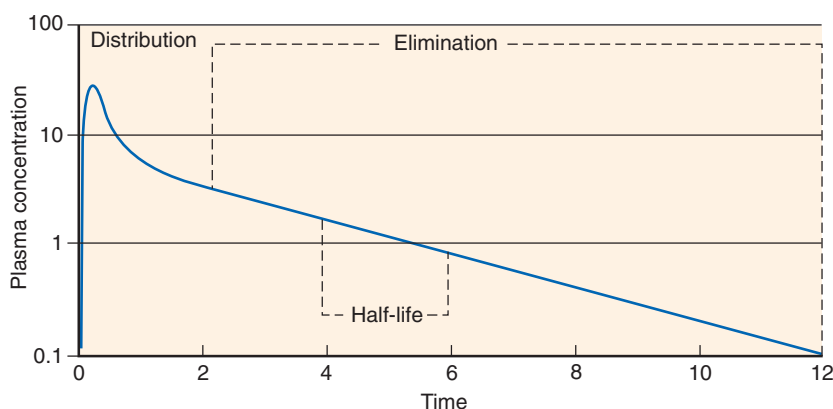


Fig. 61.1 Distribution and elimination of a drug after intravenous administration.

plasma into the extravascular space and beyond. During the terminal elimination phase, drug concentrations in plasma are in equilibrium with concentrations in body tissues (Fig. 61.1). The rate and extent of drug absorption and distribution and rate of drug elimination may be ascertained by mathematical analysis of the serum or plasma concentration data collected over an appropriate time interval. The terminal elimination half-life of a drug is the time required for the plasma concentration to decline by 50%; this can be determined from the slope during the elimination phase of the plot of serum or plasma drug concentration versus time after the drug is ingested or injected. By comparing PK data from patients with normal kidney function with PK data from patients with impaired kidney function, rational drug dosing regimens may be proposed.^{21,22}

ABSORPTION

Drugs given IV enter the central circulation directly and generally have a rapid onset of action. Drugs given by other routes must first pass through important organs of elimination before entering the systemic circulation; thus a smaller proportion of the drug reaches the systemic circulation. In many cases, only a fraction of the administered dose may reach the circulation and become available at the site of drug action. Even drugs given IV and by inhalation must pass through the lungs before reaching arterial blood. Similar to other organs, the lungs remove substantial amounts of some agents (e.g., removal of IV administered adenosine by passage through the pulmonary circulation). For drugs administered orally, the rate and extent of gastrointestinal (GI) absorption are important considerations. Absorption has been characterized by determining the maximum attained serum or plasma concentration (C_{max}), as well as the time after ingestion when the C_{max} was observed (T_{max}). Differences in these two parameters among patient groups were historically considered evidence of altered GI absorption when actually the bioavailability may have been unchanged.³⁴ The bioavailability of a drug depends on the extent of metabolism during its first pass through the intestinal wall and liver before reaching the systemic circulation. The absolute bioavailability is determined by comparing the area under the serum/plasma concentration–time curve (AUC) after oral administration with that observed after IV administration. When this measure

of bioavailability was assessed, there were very few drugs shown to be affected by the presence of CKD or AKI.^{7,35}

First-pass biotransformation may also occur in the gut; bioflavonoids in grapefruit juice can inhibit cytochrome P450 (CYP) 3A4 and noncompetitively inhibit the metabolism of drugs metabolized by this enzyme. This grapefruit juice–CYP3A4 interaction was first noted with the calcium channel blocker felodipine.³⁶ This interaction also increases the bioavailability of cyclosporine by as much as 20%.³⁷ A wide variety of other drugs are similarly affected, including several medications used for depression and anxiety (e.g., selective serotonin reuptake inhibitors, serotonin–norepinephrine reuptake inhibitors) and statins.³⁸ Herbal medicine (e.g., hypericin, one of the constituents of St. John's wort) can activate the adenosine triphosphate-binding cassette transporter or P-glycoprotein (multidrug resistance) transporter in gut mucosa, leading to reduced drug absorption.³⁹

Although GI symptoms are common in patients with ESKD, little specific information about alimentary function is available. The salivary concentration of urea increases when urea accumulates in plasma. Ammonia forms from urea in the presence of gastric urease and buffers gastric acid, increasing gastric pH. The ammonia is absorbed and converted to urea again by the liver. The gastric alkalinizing effect of this internal urea–ammonia cycle decreases the absorption of drugs that are best absorbed in an acidic environment. Drug malabsorption may be further aggravated by the increased use of various therapies to reduce gastric acidity and/or reduce phosphate absorption, especially in patients who are dialysis dependent.^{40,41} The resultant chelation and formation of nonabsorbable complexes reduce the bioavailability of some drugs, including several antibiotics and digoxin.

The processes of GI drug absorption are complex, may be saturable and dose dependent, and are more variable in patients with ESKD than in those with normal kidney function. Gastroparesis, commonly observed in patients with diabetes mellitus, many of whom also have CKD, prolongs gastric emptying and delays drug absorption (i.e., T_{max} is observed to be delayed). Conversely, diarrhea decreases gut transit time (T_{max} is shortened and diminishes drug absorption by the small bowel). Gut mucosal integrity becomes impaired across the spectrum of CKD, as evidenced by increasing levels of circulating translocated endotoxins.⁴²

DISTRIBUTION

The volume of distribution (V_D) of a drug does not necessarily correspond to a specific anatomic space. Rather, the V_D is a mathematical construct based on the plasma concentration achieved following the IV administration of a given dose of a drug. Agents that are highly protein bound and those that are water soluble tend to be restricted to the vascular compartment and extracellular fluid (ECF) space, respectively, and thus have volumes of distribution less than 0.20 L/kg. Highly lipid-soluble drugs and those extensively bound to tissues often exhibit volumes of distribution in excess of 1 L/kg. The drug distribution volume of highly water-soluble or protein-bound drugs may be increased in patients with AKI or CKD if edema and/or ascites is present (Table 61.1).^{5,7,35,43} Drug distribution is one of the most important and complicated factors to quantify in patients with AKI. There is a fine balance between detrimental fluid overload and adequate hydration to preserve and optimize perfusion and function. Critically ill patients should be managed in a slightly negative fluid balance after initial adequate fluid resuscitation has been achieved.^{44–46} If patients are volume expanded, the administration of the usual doses of many drugs will result in inadequately low plasma concentrations.

The distribution volume of drugs may be altered by fluid removal during dialysis. Changes in body cell mass (nonfat, nonwater, nonbone mineral mass) commonly occur over time in patients on dialysis, resulting in sarcopenia.⁴⁷ Failure to detect a reduction in body cell mass may lead to inappropriate maintenance of the same dry weight and drug dosage regimen, despite a real increase in total body water (and thus the distribution volume of several drugs).⁴⁸ Finally, the method used to calculate the V_D may be influenced by impaired kidney function. The three most commonly used V_D terms are volume of the central compartment (V_c), volume of the terminal phase (V_β and V_{area}), and V_D at steady state (V_{ss}). The V_c for many drugs approximates ECF volume and thus may be increased or decreased by acute changes. Oliguric acute renal failure is often accompanied by fluid overload and a resultant increased V_c for many drugs. The V_{area} or V_β represents the proportionality constant between plasma concentrations in the terminal elimination phase and the amount of drug remaining in the body. V_β is affected by distribution characteristics and by the terminal elimination rate constant. V_β and V_{ss} will often be similar in magnitude, with V_β being slightly larger. Because V_{ss} has the advantage of being independent of drug elimination, it is the most appropriate volume term to use when it is desirable to compare drug distribution volumes between patients with renal insufficiency and those with normal renal function.⁴⁹

Alterations of plasma protein binding in patients with CKD can also affect drug distribution and elimination. The V_D of a drug, quantity of unbound drug available for action, and the degree to which the agent is eliminated by hepatic or renal excretion are all influenced by protein binding. Drugs that are protein bound attach reversibly to albumin or α_1 -glycoprotein in plasma (Fig. 61.2). Whereas organic acids bind to a single binding site, organic bases probably have multiple sites of attachment.^{50–53}

Protein-bound organic acids such as hippuric acid, indoxyl sulfate, and 3-carboxy-4-methyl-5-propyl-2-furanpropionic acid accumulate in advanced CKD and decrease the

Table 61.1 Volume of Distribution of Selected Drugs in Patients With Normal Kidney Function and Those on Dialysis

Drug	Normal (L/kg)	Stage 5	Change From Normal (%)
		Chronic Kidney Disease (L/kg)	
Increased			
Amikacin	0.20	0.29	45
Cefazolin	0.13	0.17	31
Cefoxitin	0.16	0.26	63
Ceftriaxone	0.28	0.48	71
Cefuroxime	0.20	0.26	30
Doripenem	0.25	0.47	88
Dicloxacillin	0.08	0.18	125
Erythromycin	0.57	1.09	91
Furosemide	0.11	0.18	64
Gentamicin	0.20	0.32	60
Isoniazid	0.6	0.8	33
Minoxidil	2.6	4.9	88
Naproxen	0.12	0.17	42
Phenytoin	0.64	1.4	119
Trimethoprim	1.36	1.83	35
Vancomycin	0.64	0.85	33
Decreased			
Atenolol	1.2	0.9	−25
Chloramphenicol	0.87	0.60	−31
Ciprofloxacin	2.5	1.95	−22
Digoxin	7.3	4.0	−45
Ethambutol	3.7	1.6	−57
Methicillin	0.45	0.3	−33
Metoprolol	5.6	1.0	−82
Pindolol	2.1	1.1	−48
Propranolol	4.4	3.6	−18

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bombardieri A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy* 2009;29:562–577; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Murphy JE. *Clinical Pharmacokinetics Pocket Reference*. 5th ed. Bethesda, MD: American Society of Health-System Pharmacists; 2015; Verbeeck RK, Musuamba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*. 2009;65:757–773; and Olyaei AJ, Steffl JL. A quantitative approach to drug dosing in chronic kidney disease. *Blood Purif*. 2011;31:138–145.

protein binding of many acidic drugs.^{53,54} A combination of decreased serum albumin concentration and reduction in albumin affinity for the drug reduces protein binding in dialysis-dependent patients. Even when the plasma albumin concentration is normal, the protein-binding defect of

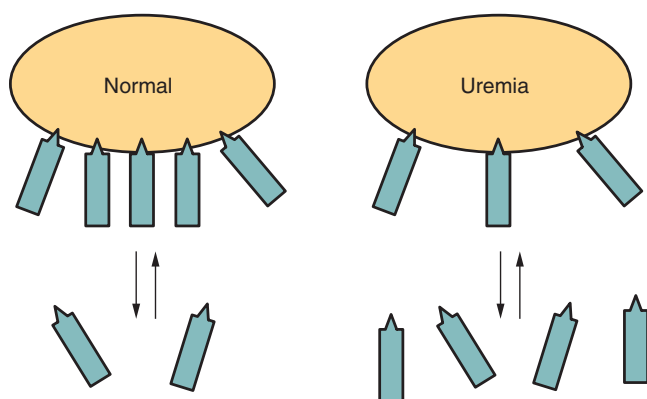


Fig. 61.2 Protein-binding defect in uremia. Displacement of the drug from its binding site by an accumulation of undefined uremic toxins or a uremia-induced conformational change in the binding-site geometry results in more free drug in the plasma.

some drugs correlates directly with the level of azotemia and may be corrected with dialysis.^{5,7,55} Binding affinity is influenced by changes in the structural orientation of the albumin molecule or by the accumulation of endogenous inhibitors of protein binding that compete with drugs for their binding sites.⁵⁰

The unbound fraction of several acidic drugs is increased in CKD because of impaired plasma protein binding. Toxicity can occur if the total plasma concentration of these drugs is pushed into the therapeutic range by increasing the dose, wherein the free (active) concentration may be in the supratherapeutic range. For such drugs, unbound plasma concentrations should be measured to guide therapy. The need to measure unbound drug concentrations applies especially to drugs with very narrow therapeutic ranges, such as phenytoin.⁵⁵ Predicting the clinical consequences of altered protein binding is difficult. Although decreased binding results in more unbound drug being available at the site of drug action or toxicity, the distribution volume is increased, resulting in lower plasma concentrations after a given dose. More unbound drug is available for metabolism and excretion, which increases the clearance and decreases the half-life of the drug in the body. Drugs with decreased protein binding in patients on dialysis are listed in Table 61.2.

METABOLISM

The disposition of drugs metabolized by the liver may be altered by changes in plasma protein binding. The systemic clearance of a highly protein-bound drug with a low hepatic extraction ratio depends on the simultaneous effects of AKI or CKD on protein binding and intrinsic metabolic drug clearance. Because the effects of severe CKD on these two factors offset each other in terms of total systemic clearance, the lowest total systemic clearance is not seen in patients with ESKD but rather occurs in patients with moderate to severe CKD. The systemic clearance of drugs with a high hepatic extraction ratio is not thought to be as susceptible to the effect of CKD as that of drugs with a low extraction ratio.⁵⁶

Table 61.2 Unbound Fraction of Selected Drugs in Patients With Normal Kidney Function and End-Stage Kidney Disease (ESKD)

Drug	Normal Patient	ESKD Patient	Change From Normal (%)
Acidic Drugs			
Abecarnil	4	15	275
Azlocillin	62.5	75	20
Cefazolin	16	29	81
Cefoxitin	27	59	119
Ceftriaxone	10	20	100
Clofibrate	3	9	200
Dicloxacillin	3	9	200
Diflunisal	12	44	267
Doxycycline	12	28	133
Furosemide	4	6	50
Methotrexate	57.2	63.8	12
Metolazone	5	10	100
Moxalactam	48	64	33
Pentobarbital	34	41	21
Phenytoin	10	21.5	115
Salicylate	8	20	150
Sulfamethoxazole	34	58	71
Valproic acid	8	23	188
Warfarin	1	2	100
Basic Drugs			
Decreased			
Bepidil	0.3	0.1	-67
Clonidine	55.6	47.6	-14
Disopyramide	32	28	-13
Propafenone	3.4	2.4	-29
Increased			
Amphotericin B	3.5	4.1	17
Chloramphenicol	45	64	42
Clonazepam	13.9	16	15
Diazepam	2	8	300
Fluoxetine	5.5	6.5	18
Ketoconazole	1	1.5	50
Prazosin	6	10.1	68
Rosiglitazone	0.16	0.22	38
Triamterene	19	43	126

Data from Matzke GR, Nolin TN. Drug dosing in renal disease.

In: Bombardieri A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed.

Philadelphia: Elsevier; 2017; Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations

for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis.

Pharmacotherapy 2009;29:562-577; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of

dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York:

McGraw-Hill; 2011; Murphy JE. *Clinical Pharmacokinetics Pocket Reference*. 5th ed. Bethesda, MD: American Society of Health-System Pharmacists; 2015; and Verbeeck RK,

Musuamba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*.

2009;65:757-773.

Table 61.3 Drugs With Pharmacologically Active Metabolites That May Affect Efficacy or Toxicity in Patients With Severe Chronic Kidney Disease

Parent Drug	Metabolite	Pharmacologic Activity of Metabolites
Acetaminophen	<i>N</i> -Acetyl- <i>p</i> -benzoquinoneimine	Responsible for hepatotoxicity
Allopurinol	Oxipurinol	Metabolite primarily responsible for suppression of xanthine oxidase
Azathioprine	Mercaptopurine	All immunosuppressive activity resides in the metabolite
Cefotaxime	Desacetyl cefotaxime	Similar antimicrobial spectrum, but 10% to 25% as potent
Chlorpropamide	2-Hydroxychlorpropamide	Similar in vitro insulin-releasing activity
Clofibrate	Chlorophenoxyisobutyric acid	Primarily responsible for hypolipidemic effect and direct muscle toxicity
Codeine	Morphine-6-glucuronide	Possibly more active than parent compound; may contribute to prolonged narcotic effect in renal failure patients
Imipramine	Desmethylinipramine	Similar antidepressant activity
Ketoprofen	Ketoprofen glucuronide	Accumulation of acyl glucuronide may worsen toxic effects (GI disturbances, impairment of kidney function)
Meperidine	Normeperidine	Less analgesic activity than parent, but more central nervous system stimulatory effects, epileptogenic
Morphine	Morphine-6-glucuronide	Possibly more active than parent compound; may contribute to prolonged narcotic effect in ESKD
Mycophenolic acid	Mycophenolic acid glucuronide	Lacks pharmacologic activity but may be associated with dose-limiting (GI) side effects
Procainamide	<i>N</i> -Acetyl procainamide	Distinct antiarrhythmic activity; mechanism different from that of parent compound
Sulfonamides	Acetylated metabolites	Devoid of antibacterial activity; elevated concentrations associated with increased toxicity
Theophylline	1,3-Dimethyl uric acid	Cardiotoxicity has been demonstrated
Zidovudine	Zidovudine triphosphate	Primarily responsible for antiretroviral activity

ESKD, End-stage kidney disease; GI, gastrointestinal.

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bombardieri A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Naud J, Nolin TD, Leblond FA, et al. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52:10S–22S; and Batistella M, Matzke GR. Drug dosing in renal failure. In: DiPiro J, Talbert R, Yee G, et al, eds. *Pharmacotherapy: A Pathophysiologic Approach*. 10th ed. New York: McGraw-Hill; 2017.

Many active or toxic metabolites depend on the kidneys for their removal from the body. The accumulation of these metabolites in patients with impaired kidney function (AKI and CKD) can explain, in part, the high incidence of adverse drug reactions in this patient population. For example, although the liver usually rapidly metabolizes morphine, it is categorized to be excreted mainly in the urine because its active metabolites, morphine-3-glucuronide (M3G) and morphine-6-glucuronide (M6G), readily cross the blood–brain barrier and bind to opiate receptors, exerting strong analgesic effects. In patients with CKD, morphine itself is metabolized more slowly, and these active metabolites increase, making prolonged narcosis and respiratory depression more likely.^{57,58} Similarly, the biotransformation of meperidine results in the production of normeperidine, a more polar metabolite that is normally rapidly excreted in the urine. Normeperidine has little to no analgesic activity but lowers the seizure threshold. In patients with impaired kidney function, repeated doses of meperidine may result in the accumulation of this potentially toxic metabolite, with resultant seizures.⁵⁹ Table 61.3 lists some drugs that form active or toxic metabolites in CKD patients and have been associated with adverse outcomes.

ALTERATIONS OF CYTOCHROME P450 ENZYME ACTIVITY

A decrease in the renal clearance (Cl_R) of drugs in patients with CKD is well appreciated. However, there is now preclinical and emerging clinical evidence suggesting that advanced CKD (stages 4 and 5) may lead to reductions in the nonrenal clearance of many medications as the result of alterations in the activities of uptake and efflux transporters, as well as CYP enzymes, in the liver and other organs (Table 61.4).^{60–62} The effect(s) of AKI and CKD on nonrenal drug clearance appear to depend on whether the reduction in renal function is acute or chronic in nature—and based on recent evidence the dialysis modality (i.e., HD or peritoneal).⁶³

Preservation of nonrenal metabolic clearance has been observed early in the course of AKI,^{64–66} and thus drug dosing schemes extrapolated from those with stable CKD may therefore result in ineffectively low drug concentrations. Furthermore, failure to appreciate that changes in serum creatinine (sCr) levels are not an accurate marker of the GFR early in AKI may lead to further dosing errors. The first reports of nonrenal clearance of drugs being affected by AKI came from the observation that the residual nonrenal clearances of vancomycin, ceftriaxone, meropenem, and

Table 61.4 Major Pathways of Nonrenal Drug Clearance (Cl_{NR})

Cl _{NR} Pathway	Selected Substrates
Oxidative Enzymes	
CYP1A2	Polycyclic aromatic hydrocarbons, caffeine, imipramine, theophylline
CYP2A6	Coumarin
CYP2B6	Nicotine, bupropion
CYP2C8	Retinoids, paclitaxel, repaglinide
CYP2C9	Celecoxib, diclofenac, flurbiprofen, indomethacin, ibuprofen, losartan, phenytoin, tolbutamide, S-warfarin
CYP2C19	Diazepam, S-mephenytoin, omeprazole
CYP2D6	Codeine, debrisoquine, desipramine, dextromethorphan, fluoxetine, paroxetine, duloxetine, nortriptyline, haloperidol, metoprolol, propranolol
CYP2E1	Ethanol, acetaminophen, chlorzoxazone, nitrosamines
CYP3A4/5	Alprazolam, midazolam, cyclosporine, tacrolimus, nifedipine, felodipine, diltiazem, verapamil, fluconazole, ketoconazole, itraconazole, erythromycin, lovastatin, simvastatin, cisapride, terfenadine
Conjugative Enzymes	
UGT	Acetaminophen, morphine, lorazepam, oxazepam, naproxen, ketoprofen, irinotecan, bilirubin
NAT	Dapsone, hydralazine, isoniazid, procainamide
Transporters	
OATP1A2	Bile salts, statins, fexofenadine, methotrexate, digoxin, levofloxacin
OATP1B1	Bile salts, statins, fexofenadine, repaglinide, valsartan, olmesartan, irinotecan, bosentan
OATP1B3	Bile salts, statins, fexofenadine, telmisartan, valsartan, olmesartan, digoxin
OATP2B1	Statins, fexofenadine, glyburide
PGP	Digoxin, fexofenadine, loperamide, irinotecan, doxorubicin, vinblastine, paclitaxel, erythromycin
MRP2	Methotrexate, etoposide, mitoxantrone, valsartan, olmesartan
MRP3	Methotrexate, fexofenadine

Data from Naud J, Nolin TD, Leblond FA, et al. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52:10S–22S; Yeung CK, Shen DD, Thummel KE, et al. Effects of chronic kidney disease and uremia on hepatic drug metabolism and transport. *Kidney Int*. 2014;85:522–528; Kagaya H, Niioka T, Saito M, et al. Effect of hepatic drug transporter polymorphisms on the pharmacokinetics of mycophenolic acid in patients with severe renal dysfunction before renal transplantation. *Xenobiotica* 2017;47:916–922; Thomson BK, Nolin TD, Velenosi TJ, et al. Effect of CKD and dialysis modality on exposure to drugs cleared by nonrenal mechanisms. *Am J Kidney Dis*. 2015;65:574–582; Joy MS, Frye RF, Nolin TD, et al. In vivo alterations in drug metabolism and transport pathways in patients with chronic kidney diseases. *Pharmacotherapy* 2014;34:114–122; Lee W, Kim RB. Transporters and renal drug elimination. *Annu Rev Pharmacol Toxicol*. 2004;44:137–166; Masereeuw R, Russel FGM. Therapeutic implications of renal anionic drug transporters. *Pharmacol Ther*. 2010;126:200–216; and Hsueh CH, Yoshida K, Zhao P, et al. Identification and quantitative assessment of uremic solutes as inhibitors of renal organic anion transporters, OAT1 and OAT3. *Mol Pharm*. 2016;13:3130–3140.

imipenem were higher in patients with AKI compared with patients with CKD, who had comparable CrCl.^{64–67}

Most of the direct evidence on metabolism in the presence of AKI has been derived from investigations in animal models. A number of drugs have been studied in a variety of AKI models. AKI is a heterogeneous insult that is often part of multisystem failure of cellular respiration, which can have protean consequences.^{68–71} CYP enzymes are affected by AKI, and the extent of these effects may depend on the mechanism of experimental AKI. Definitive conclusions on the PK of metabolized medications in AKI remain hampered by the clinical complexity and potential confounders (hypoxia, decreased protein synthesis, competitive inhibition from concomitant medications, and decreased hepatic perfusion).

In humans with CKD, the activities of CYPs appear to be relatively unaffected.^{60,72} It has been reported that CYP3A4 activity is reduced,⁶⁰ but recent studies have indicated that organic anion transporting polypeptide uptake activity is decreased.⁷² Thus, the perceived changes in CYP3A4 activity

were likely due to altered transporter activity, not to an alteration in CYP activity. The reduction of nonrenal clearance of several drugs that exhibit overlapping CYP and transporter substrate specificity in patients with stage 4 or 5 CKD supports this premise. These studies must be interpreted with caution, however, because concurrent drug intake, age, smoking status, and alcohol intake were often not taken into consideration. Furthermore, pharmacogenetic variations in drug-metabolizing enzymes that may have been present in the individual before the onset of AKI or CKD must also be considered.

RENAL EXCRETION

Renal clearance (Cl_R) of a drug is the composite of the GFR, tubular secretion, metabolism, and reabsorption, where f_u is the fraction of the drug unbound to plasma proteins.

$$Cl_R = (GFR \times f_u) + (Cl_{\text{secretion}} + Cl_{\text{metabolism}} - Cl_{\text{reabsorption}})$$

Drug elimination by filtration occurs by a pressure gradient, whereas tubular secretion and reabsorption are bidirectional

processes that involve carrier-mediated renal transport systems.^{60,73,74} Renal transport systems have been broadly classified on the basis of substrate selectivity into anionic and cationic renal transport systems, which are responsible for the transport of a number of organic acidic and basic drugs, respectively.^{51,60} Several drugs are actively secreted by one or more of these transporter families, including organic cationic (e.g., famotidine, trimethoprim, dopamine), organic anionic (e.g., ampicillin, cefazolin, furosemide), nucleoside (e.g., zidovudine), and P-glycoprotein transporters (e.g., digoxin, *Vinca* alkaloids, steroids).^{7,73,74} Alterations in filtration, secretion, or reabsorption secondary to CKD may have a dramatic effect on drug disposition.⁷⁵ For drugs that are primarily filtered, a reduction in GFR will result in a proportional decrease in renal drug clearance. However, recent evidence indicates that the clearance of cationic secreted drugs correlates poorly with measured GFR⁷⁶ and that uremic solutes inhibit OAT1 and OAT3 transporters and contribute to the decline in renal drug clearance in CKD patients.⁷⁵ OAT1 and 3 are multispecific exchangers or antiporters that transport predominantly anionic substrates against a concentration gradient from the blood into proximal tubule cells for subsequent elimination into the urine. The clinical impact of these findings limits the utility of eGFR for dosage regimen adjustment in CKD patients.

PHARMACOGENOMICS

Over the past 2 decades, genome-wide analyses have identified genetic variants that are associated with either a relative increased or decreased risk of several diseases.^{77,78} Most confer a low relative risk, are common alleles, and have low discriminatory and predictive values.^{79,80} The combined effect of a number of common weak variants seems to be more relevant. For example, the activity of different enzymes involved in warfarin and vitamin K metabolism has produced varying effects on anticoagulation. Genotyping data on two genes, the warfarin metabolic enzyme CYP2C9 and warfarin target enzyme vitamin K epoxide reductase complex 1 (VKORC1), confirmed that each can influence warfarin maintenance dose.⁸¹ Two genome-wide association studies also found a weak, but significant, effect of CYP4F2 on warfarin dosing.^{82,83} The presence of *CYP2C9**2 or *CYP2C9**3 variant alleles, which results in decreased metabolic enzyme activity, is associated with a significant decrease in the mean warfarin dose,⁸⁴ whereas *VKORC1* single-nucleotide polymorphisms identify *VKORC1* haplotypes accounting for a large fraction of the interindividual variation in warfarin dose.⁸⁵ The combination of both *VKORC1* and *CYP2C9* polymorphisms is associated with severe over-anticoagulation.⁸⁶ In contrast to these common but weak effect genetic variants, some DNA sequence variants are rare but have a strong effect; one example would be azathioprine. Polymorphisms in the gene that encodes thiopurine S-methyltransferase (*TPMT*) lead to reduced activity of this metabolizing enzyme, and patients with low activity (10% prevalence) or absent activity (0.3% prevalence) are at higher risk of bone marrow depression because of drug accumulation.^{87,88} The variability in how patients respond to drug treatments is clearly a consequence of alterations in PK and PD, as well as differences in their genotypes and/or phenotypes.^{78,89–94} In general, DNA sequence code variants

affect prodrug activation (i.e., clopidogrel), drug metabolism and degradation (i.e., warfarin), or predisposition to adverse drug effects (i.e., azathioprine).

The validity of phenotyping cocktails and their correlation with genotyping data are still in need of clarification.⁹⁵ Genotyping information is becoming more widely available than phenotyping data and this increases demands for a more individualized approach to pharmacotherapy. A key question now arises—Will patients be willing to pay for these tests?⁹⁶ Genotypic characterization now serves as the basis for dosing recommendations for some drugs, and >120 U.S. Food and Drug Administration (FDA)–approved drugs have pharmacogenomic information in their labeling, including fluoropyrimidines, codeine, selective serotonin reuptake inhibitors, tricyclic antidepressants, β -blockers, opiates, neuroleptics, antiarrhythmic agents, and statins.^{97–101}

However, the promise of pharmacogenomics has not always translated into improvements in patient care because of the inaccuracy of results and the complexities involved. In late 2013, the FDA approved four diagnostic, high-throughput, gene-sequencing devices, which represents a significant step forward in the ability to generate genomic information that will ultimately improve patient care.^{102,103} As Collins and Hamburg¹⁰² from the National Institutes of Health and FDA have stated, “There are many challenges ahead before personalized medicine can be considered truly embedded in health care. We need to continue to uncover variants within the genome that can be used to predict disease onset, affect progression, and modulate drug response.” Their identification of challenges to clinical utilization has been answered, in part, with the recent publication of the implementation plans of several U.S. hospitals that have launched clinical services to guide clopidogrel and warfarin.¹⁰⁴ With the increase in the number of clinical genetic tests requested by health care providers, there has been an increase in direct-to-consumer advertising of genetic tests, including at-home tests and those provided by private companies.

Although similar to direct-to-consumer advertising of prescription drugs, marketing of genetic testing raises additional concerns and considerations. These include limited knowledge among patients and health care providers regarding accuracy and reliability of available genetic tests, difficulty in interpretation of genetic testing results, lack of governmental oversight of companies offering genetic testing, and issues of privacy and confidentiality. New genomic findings need to be validated before they can be integrated into medical decision making. Physicians and other health care professionals will need support in interpreting genomic data, integrating these into clinical decision making, and applying the results to individual patients. With the right information and support, patients will be able to participate with their physicians in making more informed decisions.

An example of the complexity of individualizing drug therapy on the basis of genomic information is illustrated with the anticoagulant warfarin. Two published trials raised significant questions regarding the value of genomic data to guide the initial dosing of this agent.^{105,106} A genotype-guided approach to warfarin dosing failed to improve anticoagulation control during the first 4 weeks of treatment, according to the first of the articles.¹⁰⁵ Among 1015 patients assigned to usual care or usual care plus genotype, international normalized ratio (INR) results showed that the mean percentage of time

in the therapeutic range at 4 weeks was 45.2% in the genotype-guided group and 45.4% in the usual care group. Moreover, rates of the combined outcome of an inappropriately high INR of 4 or more, major bleeding, or thromboembolism did not differ significantly according to dosing strategy.

The second study reported conflicting results in that pharmacogenetic-based dosing was associated with a slightly but significantly higher percentage of time in the therapeutic INR range, with significantly fewer incidences of excessive anticoagulation (INR ≥ 4.0) in the genotype-guided group.¹⁰⁶ Thus, at present, the use of genomic information to guide warfarin dosing in persons with normal kidney function, which is much less than dosing for patients with CKD or AKI, remains controversial.¹⁰⁷ There are, however, published guidelines on the use of *TPMT* genotyping for azathioprine dosing in clinical practice.^{108,109} Several *TPMT* variant alleles exist, which vary in frequency among ethnic populations. In African and Asian individuals, *TPMT**3C is the most frequent deficient allele (6.5% and 2.5%, respectively), whereas in white individuals *TPMT**3A is the most frequently found variant (4.5%).¹¹⁰ *TPMT* testing before the prescription of azathioprine is one of the few examples of a pharmacogenetic test that has definitively made the transition from research into clinical practice. This testing improves safety by avoiding full-dose treatment in patients with a (partial) enzyme deficiency. The future of pharmacogenetics will be in treatment models in which patient characteristics such as CKD or AKI are combined with polymorphism information regarding relevant genes.

PHARMACODYNAMICS

The fundamental concept of PD is described by the Hill equation. This model has been extensively used to optimize the effects of many antimicrobial agents.¹¹¹ The principles are applicable to guide the dosing of medications in patients with CKD, as well as those with normal kidney function. In the patient with CKD, the concentration–time profile of many drugs is altered, so the dosage regimen predicted will likely be different than the normal regimen. This is because of the prolonged elimination half-life, which results in an increased AUC. Only rarely has there been evidence of an alteration in the PD concentration–effect relation in patients with AKI or CKD; PK changes predominantly contribute to the need for a modified dosing regimen.

The concentration (C) is the primary driving force that obligates altered dosage regimens to achieve the desired PD targets. The actual effect is a function of the maximum effect and the concentration producing the half-maximum effect. The Hill coefficient (H) is a measure of the sigmoidicity of the effect–concentration correlation:

$$E = \frac{E_{\max}}{1 + \left(\frac{CE_{50}}{C}\right)^H}$$

From this equation, the threshold concentration, which produces 5% of the maximum effect, and the ceiling concentration, which is associated with 95% of the maximum effect, can be derived. The higher the Hill coefficient, the higher the threshold concentration and the narrower is the range of lower and upper target concentrations; this is because

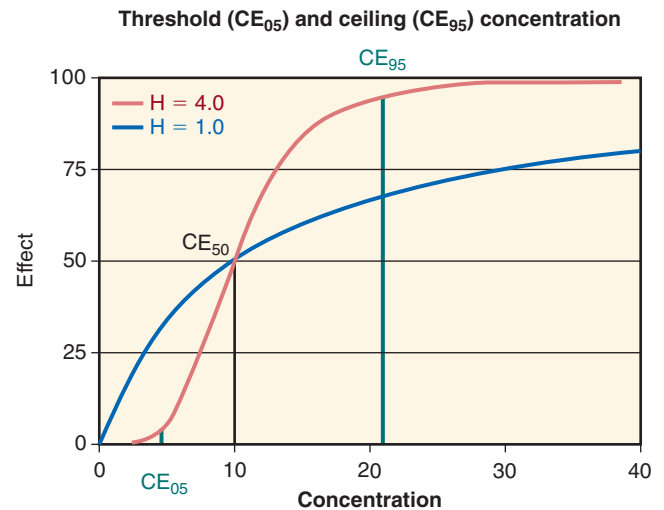


Fig. 61.3 Threshold concentration, CE_{05} , producing 5% of the maximum effect and ceiling concentration, CE_{95} , producing 95% of the maximum effect. With a Hill coefficient of $H = 1.0$, $CE_{05} = 0.5$ and $CE_{95} = 190$, whereas for $H = 4.0$, the threshold is higher, with $CE_{05} = 6.0$, but the ceiling is much less, with $CE_{95} = 21$ mg/L.

the ceiling concentration comes down close to the concentration producing the half-maximum effect (Fig. 61.3):

$$CE_{05} = 19^{\frac{-1}{H}} \cdot CE_{50}$$

$$CE_{95} = 19^{\frac{1}{H}} \cdot CE_{50}$$

The difference between the ceiling and threshold concentrations can be measured by multiples of the respective elimination half-life. The ceiling concentration is the upper limit of the targeted peak concentration ($C_{\text{peak}} < CE_{95}$), whereas the threshold concentration marks the lower limit of effective trough concentration ($C_{\text{trough}} > CE_{05}$). For a drug with a short half-life ($t_{1/2}$) and a high Hill coefficient, the therapeutic range of target concentrations can be very small (Fig. 61.3):

$$CE_{05} = CE_{95} \cdot \exp\left[-\frac{\ln(2)}{t_{1/2}} \cdot t\right]$$

$$t_{\text{ceiling-threshold}} = t_{1/2} \cdot \frac{2}{H} \cdot \frac{\ln(19)}{\ln(2)}$$

$$t_{\text{ceiling-threshold}} = t_{1/2} \cdot \frac{8.5}{H}$$

For the β -lactam ceftazidime, with a short half-life of 2.1 hours in patients with normal kidney function but with a high Hill coefficient of 3.7,¹¹² the peak to trough or ceiling to threshold time of 5 hours indicates that ceftazidime should be given at least every 6 hours to maximize efficacy. By contrast, and in agreement with the postulated postantibiotic effect, the maximum peak to trough time is estimated as 13 hours for gentamicin, with a half-life of 2 hours but a Hill coefficient of 1.3.¹¹²

The most important progress in anti-infective dosing has been achieved with the differentiation of drugs with time-dependent actions from drugs with concentration-dependent

actions.^{113,114} Specific examples are the β -lactam antibiotics and antiviral drugs with a known time-dependent effect, whereas aminoglycosides and quinolones have a concentration-dependent activity. The threshold and ceiling concentrations are specific functions of the concentration producing the half-maximum effect and the Hill coefficient. Both can be used to explain the observation that antiinfective drugs with a time-dependent effect have a significantly higher Hill coefficient than those with a concentration-dependent action.¹¹² A high Hill coefficient is associated with a high threshold concentration but, simultaneously, with a relatively low ceiling concentration. Thus, it makes no sense to increase the dose of time-dependent antiinfective drugs above the ceiling concentration. By contrast, a low Hill coefficient is associated with a high ceiling concentration and low threshold concentration. Thus it might increase the effect of concentration-dependent antiinfective drugs to give a high single dose but it is not so critical to extend the administration interval, as proposed for aminoglycosides.¹¹⁵ Practically, it is necessary to administer antiinfective drugs with a time-dependent action more frequently, whereas antiinfective drugs with a concentration-dependent action should be given with a higher maintenance dose to increase efficacy (Fig. 61.4).

Usual measures of the antimicrobial effect, such as the minimal inhibitory concentrations (MICs), AUC over MIC, time over MIC ($T > MIC$), or peak over MIC, can be unified to the following concept. The target concentration should not be less than the threshold concentration for time-dependent effects, but it could be as high as the ceiling concentration for concentration-dependent effects. A close correlation of the MIC and concentration producing the half-maximum effect has been postulated.¹¹² It was obvious, however, that for concentration-dependent antimicrobial action, the MIC could fall considerably below the concentration producing the half-maximum effect ($MIC \ll CE_{50}$). Consequently, it might be more reasonable to compare the

bacteriologic MIC with the PD parameter of a threshold concentration:

$$CE_{\text{threshold}} = CE_{05} = MIC$$

From the Hill coefficient, one can postulate that the time-dependent action and concentration-dependent action are only the extreme positions of a continuum. Every drug can be considered as concentration dependent and time dependent. To overcome resistance, a higher dose might be necessary, because relative resistance can be seen in cases in which a high concentration is required to produce the half-maximum effect. The potency is the inverse concentration producing the half-maximum effect:

$$\text{Potency} = \frac{1}{CE_{50}}$$

This concept distinguishes a relative resistance from an absolute drug resistance. A pathogen with a relative resistance can be made sensitive by increasing the dose.^{116–118} Thus, for example, it has been recommended to treat severe infections with resistant strains by increasing the standard meropenem dose from 1000 mg/day, 3 times daily, to 2000 mg/day, 3 times daily,¹¹⁹ or the daptomycin dose from 4–6 mg/kg/day to >8 mg/kg/day,¹²⁰ with careful monitoring of side effects.

Clinical Relevance

Pharmacogenetic factors predicting therapeutic response and toxicity in patients with kidney disease are emerging. However, the unpredictable changes in pharmacokinetic pathways of absorption, distribution, and elimination (metabolism, transport, and excretion) of prescribed drugs observed in patients with acute kidney injury and chronic kidney disease can confound the outcomes of personalized therapeutic interventions.

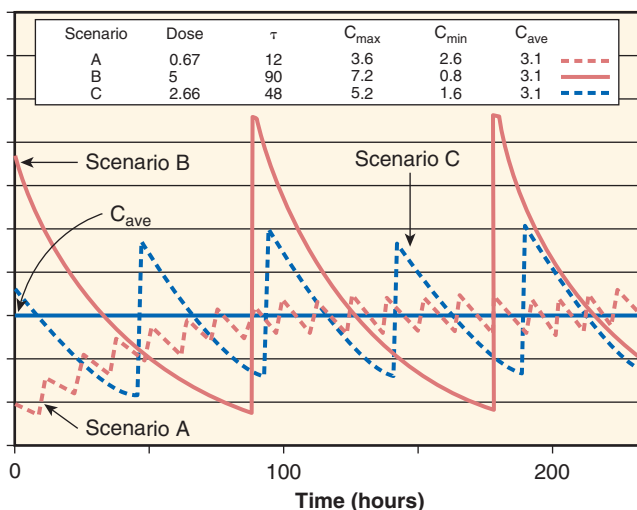


Fig. 61.4 Although the average steady-state concentrations (C_{ave}) are identical regardless of which dosage adjustment strategy one decides to use, the concentration-time profile will be markedly different if one changes the dose and maintains the dosing interval (τ) constant (Scenario A), versus changing the dosing interval and maintaining the dose constant (Scenario B), or changing both (Scenario C).

ASSESSMENT OF KIDNEY FUNCTION

The standard measure of kidney function for decades has been the GFR. GFR can be measured using many exogenous substances; however, the administration of exogenous substances is not practical for routine individual drug dose calculations in clinical practice because the procedures are not timely and not uniformly available.

CHRONIC KIDNEY DISEASE

Although GFR has been estimated based on the measured urinary clearance of creatinine (mCrCl) derived from a 24-hour urine collection, the eCrCl or eGFR (Table 61.5) is predominantly determined in clinical practice for calculation from the sCr and/or cystatin C (CysC) concentrations and patient factors.^{121–127} After years of quiescence it seems like there is a new renal function assessment tool published every few months for the past several years. The introduction of these new GFR predictive methodologies, some focused on

Table 61.5 Equations for Estimation of Creatinine Clearance or Glomerular Filtration Rate in Adults With Stable Renal Function^{a,b}

Reference	Equation
Cockcroft and Gault (1976)	Men: $\text{CrCl} = (140 - \text{age})\text{IBW}/(\text{sCr} \times 72)$ Women: $\text{CrCl} \times 0.85$
Jelliffe (1973)	Men: $\text{CrCl} = 98 - [0.8 (\text{age} - 20)]/\text{sCr}$ Women: $\text{CrCl} \times 0.9$
MDRD6 (1999)	$\text{eGFR}_{\text{Cr}} = 170 \times (\text{sCr})^{-0.999} \times (\text{age})^{-0.176} \times (0.762 \text{ if patient is female}) \times (1.180 \text{ if patient is black}) \times (\text{BUN})^{-0.170} \times (\text{Alb})^{0.318}$
MDRD4 (2000)	$\text{eGFR}_{\text{Cr}} = 186 \times (\text{sCr})^{-1.154} \times (\text{age})^{-0.203} \times (0.742 \text{ if patient is female}) \times (1.210 \text{ if patient is black})$
MDRD4-IDMS (2007)	$\text{eGFR}_{\text{Cr}} = 175 \times (\text{sCr})^{-1.154} \times (\text{age})^{-0.203} \times (0.742 \text{ if patient is female}) \times (1.210 \text{ if patient is black})$
CKD-EPI (2009)	$\text{eGFR}_{\text{Cr}} = 141 \times \min(\text{sCr}/\kappa, 1)^{\alpha} \times \max(\text{sCr}/\kappa, 1)^{-1.209} \times 0.993^{\text{age}} \times (1.018 \text{ if patient is female}) \times (1.159 \text{ if patient is black})$ • κ is 0.7 for females and 0.9 for males. • α is -0.329 for females and -0.411 for males. • min is the minimum of sCr/κ or 1. • max is the maximum of sCr/κ or 1.
Larsson et al. (2004)	$\text{eGFR}_{\text{Cys}} = 77.24 \times (\text{CysC} [\text{in mg/L}])^{-1.2623}$
Macdonald et al. (2006)	$\text{Log}_{10} \text{eGFR}_{\text{Cys}} = 2.222 + (-0.802 \times \sqrt{\text{CysC in } \frac{\text{mg}}{\text{L}}}) + (0.009876 \times \text{LM})$
CKD-EPI cystatin C Eq. (2012)	$\text{eGFR}_{\text{Cys}} = 133 \times \min(\text{sCys}/0.8, 1) - 0.499 \times \max(\text{sCys}/0.8, 1) - 1.328 \times 0.996^{\text{age}} (\times 0.932 \text{ if female})$ • sCys is serum cystatin C. • min is the minimum of $\text{sCys}/0.8$ or 1. • max indicates the maximum of $\text{sCys}/0.8$ or 1.
CKD-EPI creatinine–cystatin C Eq. (2012)	$\text{eGFR}_{\text{Cr-Cys}} = 135 \times \min(\text{sCr}/\kappa, 1)^{\alpha} \times \max(\text{sCr}/\kappa, 1) - 0.601 \times \min(\text{sCys}/0.8, 1) - 0.375 \times \max(\text{sCys}/0.8, 1) - 0.711 \times 0.995^{\text{age}} (\times 0.969 \text{ if female}) (\times 1.08 \text{ if black})$ • κ is 0.7 for females and 0.9 for males. • α is -0.248 for females and -0.207 for males. • min indicates the minimum of sCr/κ or 1. • max indicates the maximum of sCr/κ or 1.

^aFor SI conversion purposes, serum or plasma creatinine is converted from $\mu\text{mol/L}$ to mg/dL by multiplying by 0.0113; conversion from creatinine clearance conventional units of mL/min to SI units of mL/s requires multiplication by 0.0167.

^bEquations compiled from references 121–138.

Alb, Albumin; *BUN*, blood urea nitrogen; *CKD-EPI*, Chronic Kidney Disease Epidemiology Collaboration; *CrCl*, creatinine clearance in mL/min ; *CysC*, cystatin C; *eGFR*, estimated glomerular filtration rate; *IBW*, ideal body weight (kg); *IDMS*, isotope dilution mass spectroscopy; *LM*, lean mass; *MDRD*, Modification of Diet in Renal Disease; *sCr*, serum or plasma creatinine (mg/dL).

children or ethnicity, has made it challenging to ascertain their place and relative value in clinical practice. The advantages of several of these methods are that timely results are available for routine clinical practice and for many patients they provide a reasonable approximation of measured GFR or mCrCl . The variation in sCr assays, as described in the past decade, led to differences in reported sCr values among as well as within laboratories.¹²⁸ To address this issue the U.S. National Institute of Standards and Technologies released materials that are traceable to the certified reference materials for creatinine, whose roughly 5% lower values were assigned using isotope dilution mass spectroscopy (IDMS).^{122,129} It is now estimated that most laboratories currently report creatinine values traceable to this reference method. The use of IDMS creatinine assays will likely lead to less variation in kidney function estimates and theoretically more consistent drug dosing recommendations across institutions and clinical settings. Estimated GFRs based on current creatinine assays are likely to yield different CKD diagnostic categorization as well as drug dosage recommendations from those intended

by the original study, even if the same estimating equation is used due to this change in analytic methodology.¹³⁰ It is not possible or practical to repeat all the PK studies with standardized creatinine-determined eCrCl or eGFR , and therefore it is still reasonable to use drug dosing adjustments that appear in FDA- and European Medicines Agency (EMA)-approved product labeling.

Traditionally, drug dosing was based on eCrCl using the Cockcroft and Gault (CG) formula.^{5,7,18,19} For implementation in the chemical laboratory report, the CG equation is not suitable because body weight is usually not available in the electronic health record. The Modification of Diet in Renal Disease (MDRD) equations, which do not require body weight, were developed from an extensive sample of patients with CKD, all of whom had a measured GFR (i.e., iothalamate clearance) of $<90 \text{ mL/min/1.73 m}^2$.^{131,132} They were initially used by clinical laboratories, although they were only validated for patients with a $\text{GFR} < 60 \text{ mL/min}$. Therefore, the new Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation was developed to allow estimation of GFR

throughout the full range of the CKD.¹²⁵ The CKD-EPI eGFR equation has recently replaced the MDRD equation as the primary index for the staging of CKD, and values are now reported throughout the GFR range by Quest and LabCorp, the two largest laboratory service providers in the United States. For classifying kidney function into one of the five stages of CKD, the standardized CKD-EPI formula is currently preferred.¹³² Both the MDRD and CKD-EPI equations estimate the GFR for a standard 1.73-m² body surface area (BSA); thus for an individual patient, the BSA must be determined separately so that the eGFR can be expressed in milliliters per minute (mL/min).

$$\text{BSA} = \frac{\sqrt{\text{weight (kg)} \cdot \text{height (m)}}}{60}$$

$$\text{GFR} = \text{eGFR} \cdot \frac{\text{BSA}}{1.73 \text{ (m}^2\text{)}}$$

Serum cystatin C has been proposed as an alternative marker to estimate GFR, rather than sCr. Multiple equations have been proposed to estimate GFR from age, weight, gender, race, and muscle mass based on serum cystatin C measurements.¹³³ The combined use of both serum markers, cystatin C and creatinine, allows an even more accurate estimate of kidney function than either of them alone.¹³⁴ Adjusting drug doses based on the measurement of cystatin C appears to be an effective and valid tool in the limited number of applications (mainly relating to chemotherapy and antibiotic dosing) for which it has been studied.^{135–138}

The automated reporting of eGFR in the clinical setting has led some practitioners to consider substituting eGFR in place of eCrCl for renal dose adjustments. Others argue that use of the MDRD and CKD-EPI equations for drug dosing is not appropriate given that the PK studies were performed using eCrCl via the CG equation.^{18,126} Furthermore, many studies have highlighted discordance between drug dosing recommendations based on these equations.^{139–147} These studies have compared dosing recommendations based on three or more different equations for commonly used drugs in CKD patients. Average discordance rates for the MDRD Study and CG equations were between 20% and 30%.^{139–144} Another study that evaluated eight antimicrobial dosing regimens based on CKD-EPI, MDRD, and CG equations demonstrated that overall discordance rates were 15%–25% between CG and CKD-EPI and 7%–12% between MDRD and CKD-EPI.¹⁴⁵ Two recent reports highlight the challenge of choosing the best eGFR method to project aminoglycoside clearance and attainment of the desired PD endpoint.^{146,147} Major limitations with these studies include equations were not compared with a gold standard such as measured GFR and the studies did not assess drug concentrations or clinical outcomes. Although Stevens et al.¹⁴⁸ described average concordance rates for the MDRD Study and CG equations to measured GFR to be 88% and 85%, respectively; this was a simulation study and they did not evaluate drug concentrations or patient outcomes. Therefore none of these equations for estimating GFR should be used as the primary determinant for drug dosing decision making. Potential discrepancies in kidney function estimates and corresponding drug dosing regimens necessitate careful consideration of the risk–benefit ratio of each approach within the context of the complete clinical picture of the patient.

PEDIATRICS

The original equation to estimate GFR, as described by Schwartz and colleagues,¹⁴⁹ is dependent on the child's age and length:

$$\text{GFR} = (\text{length [cm]} \times k) / \text{sCr (in mg/dL)},$$

where *k* is defined by age group: infant (1 to 52 weeks) = 0.45; child (1 to 13 years) = 0.55; adolescent male = 0.7; and adolescent female = 0.55. The sCr level in μmol/L can be converted to mg/dL by multiplication using 0.0113 as the conversion factor. A newer version of the Schwartz equation¹⁵⁰ was developed from a population of 349 children (age 1–19 years) with mild to moderate CKD enrolled in the Chronic Kidney Disease in Children (CKiD) study:

$$\text{GFR} = 0.41 \times (\text{length in cm}) / \text{sCr in mg/dL}$$

Lee and associates¹⁵¹ have recently reported that this new Schwartz equation performed better than the original Schwartz equation for patients with moderate CKD, but was less accurate in patients with mild CKD. In pediatric patients, methods incorporating cystatin C have several advantages for evaluating kidney function.¹⁵² The most recent eGFR equation evaluated in pediatrics includes use of cystatin C, blood urea nitrogen (BUN), sCr level (in mg/dL) and demographic data derived from over 600 pediatric patients enrolled in the CKiD study¹⁵³:

$$\begin{aligned} \text{eGFR (mL/min/1.73 m}^2\text{)} \\ = 39.8 \times (\text{ht [m]}/\text{sCr})^{0.456} \times (1.8/\text{cystatin C})^{0.418} \\ \times (30/\text{BUN})^{0.079} \times 1.076^{\text{male}} \times (\text{ht [m]}/1.4)^{0.179} \end{aligned}$$

This equation had the highest *R*² value (0.863) and the highest frequency of values within 30% of iohexol-measured GFR (91.3%) when compared with seven other GFR estimating equations.

ACUTE KIDNEY INJURY

At present, the staging of AKI is based on sequential measurement of the sCr level and urine output.^{154–158} Because the GFR is inferred from the sCr or cystatin C, all estimates of kidney function are delayed and lag the real-time GFR. Although several methods have been proposed to estimate GFR in this patient population, none have been rigorously evaluated, and their use in clinical practice is extremely limited.^{159–162} The latest proposed method to estimate GFR in patients with AKI is the kinetic GFR (kinetGFR), which is based on age (years), weight (kg), and sCr (μmol/L) and holds true for increasing and decreasing kidney function.¹⁶³

$$\begin{aligned} \text{kinetGFR} = \frac{[150 - \text{age (years)}] \cdot \text{weight (kg)}}{\text{Cr}_1 \text{ (}\mu\text{mol/L)}} \\ \cdot \left[1 - \frac{\text{Cr}_2 - \text{Cr}_1}{t_2 - t_1} \cdot \frac{24 \text{ (hours)}}{200 \text{ (}\mu\text{mol/L)}} \right] \end{aligned}$$

This approach is based on an estimate of the creatinine production similar to the CG equation.¹²¹ The kinetic eGFR incorporates changing creatinine values over specified time intervals as well as the actually measured sCr values. It relates the increase in sCr within a specified time interval to the maximum increase in creatinine level in 1 day. Because

creatinine excretion in the urine corresponds to creatinine production, the maximum increase in sCr is about 200 $\mu\text{mol/L}$ if the patient's actual GFR is 0. The kinetic eGFR predicts what subsequently will be measurable, but in fact is already the case with kidney function. The kinetic eGFR solves the problem that there is always a delay between rapidly changing kidney function and measurable variables—namely, sCr or urine output. The calculation of a patient's kinetic eGFR may allow one to use the eCrCl- or eGFR-based dose adjustment recommendations derived from patients with CKD and be applicable in part for those with AKI.¹⁶³ Rigorous independent studies will be needed to confirm its validity and utility in clinical practice.

PATIENTS RECEIVING DIALYSIS

Some ESKD patients on dialysis as well as those with AKI on CRRT have residual kidney function that substantially contributes to better patient outcomes and the elimination of drugs and their metabolites.¹⁶⁴ Unfortunately, estimating residual kidney function in patients undergoing dialysis is challenging because the sCr concentration reflects not only residual kidney function, but also the efficiency of dialysis and role of muscle mass in creatinine generation. CrCl measurements are less reliable as a measure of GFR in patients on HD or CRRT than in those with earlier stages of CKD because of the following: (1) the volume of urine output is heavily influenced by changing effective arterial blood volume during the cyclic changes that are inherent as a result of intermittent ultrafiltration; (2) the sCr concentration changes over the duration of the clearance measurement; and (3) tubular secretion of creatinine contributes to its clearance. Estimation of residual kidney function in patients on HD or CRRT is often done by calculating the mean of a measured urea and mCrCl. Measuring the elimination of iothexol after an IV dose has been reported to be an accurate and safe measure of residual kidney function in patients on dialysis and can inform drug dosing.^{166,167} Recently Vilar et al.¹⁶⁸ and Shafi and colleagues¹⁶⁹ have reported that measurement of beta trace protein, also known as the enzyme prostaglandin H_2 D-isomerase, or β_2 -macroglobulin serum concentrations provides clinically useful estimates of residual renal function that can be used to individualize dialysis dose and perhaps could be used to individualize drug dosing.

Which one of the many eCrCl or eGFR equations should be used to determine the degree of adjustment of drug dosage regimens for patients with AKI or CKD? The pros and cons of the various GFR estimating equations are reviewed earlier. The MDRD and CKD-EPI equations significantly overestimated CrCl (mCrCl and CG) in older individuals.^{141,166} This has led to dose calculation errors for many drugs, particularly in the elderly and in individuals with severe CKD. It is an advantage of the CG equation that body weight is considered as a determinant of drug distribution volume. The choice of the optimal GFR estimating equation is of utmost importance for drugs with a narrow therapeutic index for which dosing individualization is often continuous rather than categorical. Finally, because most PK studies in patients with CKD conducted over the past 40 years have used eCrCl or mCrCl as the estimate of GFR, the CG method in adults and the latest Schwartz method in children remain the criteria to be used.¹⁸

Thus we have concluded that current evidence does not support the use of eGFR equations in place of the CG equation in older adults for the purpose of renal dosage adjustments. However, for patients with AKI, there is no obvious best choice for GFR estimation to guide drug dosing.

DRUG DOSING CONSIDERATIONS

PATIENTS WITH CHRONIC KIDNEY DISEASE

Despite the availability of numerous guidelines regarding drug dosing for patients with impaired kidney function, there is insufficient evidence as to which, if any, is preferred.^{6,19,51,170–176} Occasionally, recommendations derived from postmarketing studies conflict with the information in these reports, as well as the official FDA or EMA product labeling. Prior to 1998, there were no official guidelines regarding when and how to characterize the relationship between the PK and PD of a drug and kidney function. The FDA guidelines issued in May 1998¹⁷⁷ and the 2010 proposed revision,¹⁷⁸ and the EMA guidelines of 2015¹⁷⁹ have provided frameworks for which drugs should be evaluated and guidance regarding study design, data analysis, interpretation of study results, and recommendations for the incorporation of data into product labeling.

GOALS OF THERAPY

The desired goal is typically the maintenance of a similar peak, trough, or average steady-state drug concentration or, for antibiotics, an optimized PD measure, such as the peak over MIC, the time above the MIC, or the ratio of the drug area under the AUC to the MIC, as would be optimal for persons with normal kidney function (see the “Pharmacodynamics” section for more detail). When there is a significant relationship between drug concentration and clinical response¹¹² (e.g., aminoglycosides) or toxicity⁵⁵ (e.g., phenytoin), attainment of the specific target values becomes critical. If, however, no specific PK or PD target values have been reported, a regimen goal of attaining and maintaining the same average steady-state concentration may be appropriate.

FDA-approved drug labels, and commonly used drug information sources such as American Hospital Formulary Service Drug Information, Goodman and Gilman's *The Pharmacological Basis of Therapeutics*, the *British National Formulary*, and *Drug Prescribing in Renal Failure* are excellent sources of information about a drug's PK characteristics, goals of therapy, and the dosing regimens to achieve the goals (i.e., adequate drug concentrations or PD target values, in various patient populations).^{173–176} However, in the digital era, many clinicians rely on drug information via their smartphone or personal digital assistant. Commonly used programs that answer frequent bedside drug questions, such as the proper dosing, pharmacology, adverse side effects, drug–drug interactions, and pregnancy safety, are Epocrates Rx,¹⁸⁰ Medscape Mobile,¹⁸¹ Micromedex Drug Information,¹⁸² and mobilePDR.¹⁸³ There are more specific applications for renal drug dosing but they require payment to access the information: KidneyCalc¹⁸⁴ and ABX Dosage–Adjustments in Renal Failure.¹⁸⁵ Regardless of dosing recommendations by the various sources, the design of the optimal dosage regimen is dependent on the availability of an accurate characterization of the relationship between

the PK parameters of the drug and renal function and an accurate assessment of the patient's renal function.

INDIVIDUALIZATION OF THE DRUG DOSAGE REGIMEN

Most dosage adjustment guidelines have proposed the use of a fixed dose or interval for patients with broad ranges of kidney function.^{51,170–175} The mild, moderate, and severe CKD categories vary among reference sources, so the recommended regimen may not be “optimal” for all patients whose kidney function lies within the range, especially for agents with a narrow therapeutic index.¹⁸ The approach to developing drug dosage adjustment recommendations for the patient with CKD is predicated on attainment of the desired exposure goal at steady state. To achieve the desired goal in a timely fashion, a stepwise approach that includes multiple considerations (Table 61.6) for each individual drug should be considered.¹⁷² The following considerations may help guide individualization of therapy.

The initial or loading dose (LD), which in many patients with AKI will be larger than the typical maintenance dose, should be calculated to achieve the desired $C_{\max}$ therapeutic drug concentration. An LD should be used for most patients with stage 4 or 5 CKD to achieve the desired steady-state concentration rapidly and in which the V_D of a drug is significantly increased in extracellular volume-expanded patients with AKI and/or CKD relative to those with normal kidney function. If the relationship between V_D and CrCl has been

characterized, then the V_D should be estimated from that relationship. If no LD is prescribed, four to five half-lives of the drug will pass before the desired steady-state plasma concentration is achieved; however, doing so may contribute to therapeutic failure. The proportion of the LD given affects the magnitude of the steady-state plasma concentration and how rapidly plasma concentrations are achieved. An LD equivalent to the dose given to a patient with normal kidney function should be given to patients with impaired kidney function if the drug's half-life is especially long and if the physical examination suggests normal ECF volume. If the patient has marked volume expansion or evidence indicates that the V_D of the drug is larger in patients with CKD, then a higher dose can be calculated from the following expression:

$$LD = V_D \times C_{\max} \times \text{actual weight/IBW},$$

where V_D is the drug's volume of distribution (in liters per kilogram of IBW in those with CKD), IBW is the patient's ideal body weight (in kilograms), and $C_{\max}$ is the desired steady-state maximum plasma drug concentration.

The primary reference for information regarding the maintenance dose for patients with CKD should be the FDA and/or EMA official product labeling. If no official drug dosing guidance is available, one may need to search the literature to find a recommendation strategy derived from nonregulatory or postmarketing clinical investigations. If no such resource is found, one can consult online or published tertiary references that have developed dosing recommendations based on the Dettli or Tozer method, initially published in 1974.^{21,22} They used similar foundational PK characteristics and approaches to calculate the maintenance dose for a patient with a given eCrCl. In essence, either the dose (D) should be reduced or the interval (τ) extended. When the dose is reduced, the $C_{\max}$ will be lower and the trough concentrations will be higher than those observed in persons with normal kidney function. When the administration interval is extended, the peak and trough concentrations are kept constant but the dosing frequency decreases (Fig. 61.4).

To maintain the normal dose interval in patients with impaired kidney function, the amount of each dose after the LD can be estimated from the following equation:

$$D_f = D_n \times Q,$$

where D_f is the dose for the patient with impaired kidney function to be given at the normal dosing interval, D_n is the normal dose, and Q is the dosage adjustment factor. The dosage adjustment factor (Q) can be calculated as

$$Q = 1 - (FE[1 - KF]),$$

where FE is the fraction of the drug eliminated unchanged renally in a patient with normal renal function and KF is the ratio of the patient's CrCl or GFR to the assumed normal value of 120 mL/min (equivalent to 2.00 mL/sec). Thus, for a drug that is 85% eliminated unchanged by the kidneys, the Q factor in a patient who has a CrCl of 10 mL/min (0.17 mL/sec) would be as follows:

$$\begin{aligned} Q &= 1 - \left(0.85 \left[1 - \frac{10}{120} \right] \right) \\ &= 1 - (0.85[0.92]) \\ &= 1 - 0.78 \\ &= 0.22 \end{aligned}$$

Table 61.6 Stepwise Approach to Adjust Drug Dosage Regimens for Patients With Impaired Kidney Function

Step	Process	Assessment
1	Obtain history and relevant demographic and clinical information	Record demographic information, obtain past medical history (including history of renal disease), and record current laboratory information (e.g., serum creatinine).
2	Estimate creatinine clearance	Use Cockcroft–Gault equation to estimate creatinine clearance, or calculate creatinine clearance from timed urine collection.
3	Review current medications	Identify drugs for which individualization of the treatment regimen will be necessary
4	Calculate individualized treatment regimen	Determine treatment goals (see text); calculate dosage regimen based on pharmacokinetic characteristics of the drug and patient's renal function.
5	Monitor	Monitor parameters of drug response and toxicity; monitor drug levels if available or applicable.
6	Revise regimen	Adjust regimen based on drug response or change in patient status (including renal function), as warranted.

Adapted from Battistella M, Matzke GR. Drug dosing in renal failure. In: DiPiro J, Talbert R, Yee G, et al., eds. *Pharmacotherapy: A Pathophysiologic Approach*. 10th ed. New York: McGraw-Hill; 2017.

If one desires to give the same maintenance dose, a factor that may be required because of the limited availability of alternative formulations, the dosing interval at which the normal dose should be administered can be calculated as follows:

$$\tau_f = \tau_n / Q$$

The decision to extend the dosing interval beyond a 24-hour period should be based on the need to maintain therapeutic peak or trough levels. The dosing interval may be prolonged if the peak level is most important. Prolonging the dose interval in patients on dialysis is frequently a convenient method to modify the drug dosage regimen. This method is particularly useful for drugs with a long plasma half-life. In general, drugs removed by dialysis given once daily should be given after the dialysis treatment, with aminoglycosides being a controversial exception.^{186–190}

A third alternative that is especially helpful when the calculated dose or dosing interval is impractical is to select the administration interval according to the target trough concentration while the peak is kept constant:

$$\tau_{\text{target}} = \left(t_{1/2} / 0.693 \right) \times \ln(C_{\text{peak}} / C_{\text{trough-target}})$$

$$D = LD \times (1 - C_{\text{trough-target}} / C_{\text{peak}})$$

Alternatively, one can calculate the adjusted dose (D_p) to be given at the predetermined practical dosage interval (τ_p or τ_{ptarget}) as follows:

$$D_p = (D_n \times \tau_p \times Q) / \tau_n,$$

where τ_p is the estimated dosing interval, as calculated from the aforementioned equation for τ_{ptarget} , or the clinically practical value for the renally impaired patient (e.g., 12, 18, 24, 36, and 48 hours). These approaches, which use a combination of the dose reduction and interval prolongation methods, are often the most clinically practical. When in doubt, clinicians should consult an experienced pharmacist, preferably one with extensive experience in evaluating patients with CKD and altered body composition (e.g., fluid overload).

MEASUREMENT OF THERAPEUTIC DRUG LEVELS

Measuring drug concentrations is one way to optimize therapeutic regimens and account for changes among and within individuals. Therapeutic drug monitoring requires availability of rapid, specific, and reliable assays and known correlations of drug concentration to therapeutic and toxic outcomes. Indications for therapeutic drug monitoring include an experimentally determined relationship between plasma drug concentration and the pharmacologic effect, medications with a narrow therapeutic window, knowledge of the drug level influences management, potential patient compliance problems, and the drug dose cannot be optimized by clinical observation alone. Examples of drugs analyzed by therapeutic drug monitoring include digoxin, lithium, phenytoin, theophylline, valproic acid, warfarin, tacrolimus, sirolimus, aminoglycosides, vancomycin, and clozapine. Unless therapeutic drug monitoring is being used to forecast a dose or there are concerns about toxicity, samples should be taken at steady state (four to five half-lives after starting therapy). The timing of the collection of the sample is important as the drug concentration changes during the dosing interval. The least variable point in the dosing interval to collect a

blood sample is just before the next dose is due. The drug concentration is complementary to and not a substitute for clinical judgment, so it is important to treat the individual patient and not the laboratory value. Drug concentrations may be used as surrogates for drug effects so therapeutic drug monitoring may assist with dose individualization. It can also be used to detect toxicity, so therapeutic drug monitoring can optimize patient management and improve clinical outcomes.¹⁹¹

Hypoalbuminemia may influence interpretation of drug concentrations because the total drug concentration may be reduced, even when the active unbound drug concentration generally is not. Unbound drug concentrations are often not clinically available, so clinicians must empirically consider the influence of hypoalbuminemia in their interpretation of measured total drug concentrations, as in the case of phenytoin and several antibiotics (e.g., ceftriaxone, daptomycin).^{55,192,193}

PATIENTS WITH ACUTE KIDNEY INJURY

Critically ill patients frequently develop AKI, depending on the definition, from 5% to 15% of all non–same-day hospitalization care is complicated by AKI.^{9–13} In most cases, drug dosing is based on drug disposition information derived from studies in stable patients with CKD. Unfortunately, there are large gaps in knowledge of drug metabolism and disposition in patients with AKI; thus patients may be at significant risk for underdosing as well as overdosing. More than 30 definitions of AKI have been published in the literature.^{155–158} The lack of a consensus definition and classification of AKI reflects the wide range of causes and severity with which it presents. The presentation can vary from part of multiorgan dysfunction in critically ill patients to isolated AKI.¹⁹⁴ As a result, AKI-related, in-hospital mortality rates vary from 70% in intensive care unit (ICU) patients¹⁹⁵ to 35% in other hospitalized patients.¹⁹⁶

The potential effects of AKI on drug dosing are of major consequence because AKI patients are often critically ill and require multiple drug therapies, some of which may be nephrotoxic or require dose modification in the setting of AKI. The PK changes in absorption, distribution, metabolism, and excretion presented earlier in this chapter, and in other sources, are foundational to optimal patient care.^{7,51,172,197,198} The clinician needs to appreciate these factors and realize that they may worsen and improve over the period of evolution or recovery of the AKI episode. Critically ill patients with AKI typically have minimal oral intake of food and liquids and commonly require parenteral administration of drugs otherwise given orally (e.g., antihypertensives, immunosuppressives).

There is a paucity of dosing algorithms to guide pharmacotherapy, derived from investigations of the PK and PD of multiple dose studies in patients with AKI. Most of the critical care literature and almost all FDA or EMA product labeling contain drug dosage recommendations derived from observations of patients with CKD and ESKD. The limited data available in the setting of AKI have predominantly been developed by clinicians; rarely is this information incorporated into official product labeling. The principles of drug dosage regimen modification described earlier for use in CKD thus remain the foundation for therapy optimization in patients with AKI.

LOADING DOSE

Many patients with AKI are overhydrated, and the distribution volume is much larger than under normal conditions. Thus, the LD may need to be higher than the normal starting dose for persons with normal kidney function. Because the V_D of many drugs, especially hydrophilic antibiotics, including β -lactams, cephalosporins, and carbapenems, is significantly increased in the presence of AKI, the administration of proactive LDs (25% > normal) is highly recommended.

MAINTENANCE DOSE

Forecasting the degree and rate of change in kidney function and fluid volume status is extremely challenging. Thus maintenance dosing regimens for many drugs, especially antimicrobial agents, should be initiated at normal or near-normal dosage regimens and adjustments made based on the relationship between drug PK characteristics and kidney function, as described earlier. Prospective measurement of serum drug concentrations and analysis using state-of-the-art PK and PD approaches should be used whenever possible.

PATIENTS UNDERGOING HEMODIALYSIS

The optimization of pharmacotherapy for patients receiving maintenance HD and emergent HD are both critically dependent on the availability of reliable information from well-designed PK studies.^{7,17,25,199–204} The impact of HD on drug therapy is dependent on the drug characteristics and dialysis prescription. Drug-related factors include molecular weight (MW) or size, degree of protein binding, and distribution volume.^{172,201} The vast majority of HD filters in use up until the mid-1990s was generally impermeable to drugs with an MW > 1 kDa.^{199–201} Dialysis membranes in the 21st century are predominantly composed of semisynthetic or synthetic materials, which have larger pore sizes, and this allows the ready passage of drugs that have an MW up to 20 kDa.

Drug clearance during dialysis can occur by three different processes.^{7,201} Drug removal by conventional HD occurs primarily by diffusion down a concentration gradient from the plasma to the dialysate. Removal of low-MW drugs is enhanced by increasing blood and dialysate flow rates and by using large-surface-area dialyzers. Larger molecules require more porous membranes for increased removal. The clearance of a drug by conventional HD can be estimated from the unbound fraction (f_u) and the following relationship:

$$Cl_{HD} = f_u \times Cl_{urea} \times (60/MW_{drug}),$$

where Cl_{HD} is the drug's clearance by HD, Cl_{urea} is the dialyzer clearance of urea, and MW_{drug} is the MW of the drug. The urea clearance for most conventional dialyzers varies between 150 and 200 mL/min and is comparable with values reported with high-flux hemodialyzers.²⁰¹ With high-flux HD, the V_D and degree of protein binding of the drug become more important determinants of dialyzer clearance. The hemodialyzer clearance of drugs that are not highly protein bound and have relatively small volumes of distribution runs in parallel to urea clearance, despite their large molecular mass.²⁰⁵ The convective transport and removal of drugs during high-flux HD

depends primarily on filtration pressure gradient, treatment time, blood, and dialysate flow rates. Despite the widespread adoption of high-flux HD in certain parts of the world, there are sparse quantitative data on drug clearance.

Total daily small solute removal is more efficient if the frequency of HD is increased. Daily and nocturnal dialysis therapies yield different clearance values compared with thrice-weekly, high-flux, in-center HD, and also differ from each other. Although there is an increase in dialysis hours, which would suggest an increase in drug removal, the blood and dialysate flow rates are slower and thus drug clearance per unit of time is less. There has been little investigation of the effects of sustained low-efficiency hemodialysis (SLED) regimens on drug disposition or comparison among modalities.^{206–209} Only a few agents have been evaluated during the delivery of one of these dialytic variants. Slow nocturnal dialysis required a significant increase in gentamicin dosage to achieve therapeutic levels compared with conventional thrice-weekly dialysis.^{206,207} The variability in drug clearance was high and did not correlate with small solute clearance. Similar findings were reported with cefazolin.²⁰⁸ The cefazolin clearance during NHD was slightly lower ($Cl = 1.65$ L/hr) than during high-flux intermittent HD ($Cl = 1.85$ L/hr). However, a greater percentage of cefazolin was removed in 8 hours of NHD (80%) than conventional 4-hour high-flux HD (60%). The investigators concluded that a dosing regimen of a 2-g LD followed by 1-g IV dose after each NHD was sufficient to achieve concentrations $6 \times MIC$ for *Staphylococcus* species for at least 70% of the dosing interval. As a result, though drug dosing in patients receiving one of these dialytic variants may be empirically increased, drug dosing regimens should be guided by drug-level monitoring when feasible. Drugs with a molecular size of 500 to 5000 Da appear to be particularly likely to have an increased clearance with this modality. Studies of modeled clearance have suggested that frequent HD regimens would be associated with enhanced clearance (and thereby the potential of under dosing) of daptomycin.^{210,211} This enhanced clearance was confirmed in the setting of AKI when the PK associated with extended daily dialysis (EDD) was investigated.¹⁹² These findings should be transferable to maintenance HD, with a degree of caution about the effects on distribution volumes that might arise in the setting of septic shock.²¹² One of the other effects of prolonged HD appears to be a reduction in rebound of drug concentrations after the termination of dialysis.^{213,214} This is probably because the rate of transfer from the peripheral to central compartment relative to the rate of diffusive removal is lower.

There were >100 different dialysis or hemofilters available in the United States in 2013, and at least four distinct variants of HD are currently being used.²⁰¹ The effect of HD or hemofiltration on the disposition of a drug may vary markedly and, because dialyzer or hemofilter clearance is rarely evaluated more than once, clinicians have to extrapolate data from one procedure to another.^{215–217} The enhanced efficiency of 21st century dialyzers means that most of the literature for medications developed prior to 2000 probably reflects an underestimation of the impact of HD.²⁰⁰ Consequently, the dosage may need to be empirically increased by 25% to 50%. Therapeutic drug monitoring should be used for drugs with narrow therapeutic indices to optimize safety and efficacy.

ASSESSMENT OF THE IMPACT OF HEMODIALYSIS

The most commonly used means for assessing the effect of HD is to calculate the dialyzer clearance of a drug (Cl^p_D) from plasma, as follows:

$$Cl^p_D = Q_p([A_p - V_p]/A_p),$$

where Q_p is plasma flow through the dialyzer, A_p is the concentration of drug in plasma going into the dialyzer, and V_p is the plasma concentration of drug leaving the dialyzer.^{199,214} This equation tends to underestimate HD clearance for drugs that readily partition into and out of erythrocytes. In addition, venous plasma concentrations may be artificially high if extensive ultrafiltration is performed, so Cl^p_D will be lower than it really is. Because of these limitations, the recovery clearance approach remains the benchmark for the determination of dialyzer clearance and can be calculated as follows^{175,199–201}:

$$Cl^r_D = R/AUC_{0-t},$$

where R is the total amount of drug recovered unchanged in the dialysate and AUC_{0-t} is the area under the predialysis plasma concentration–time curve during the period that the dialysate was collected. The HD clearance values reported in the literature may vary significantly, depending on which of these methods were used.²⁰⁰

It is common practice in most HD units to administer drugs after dialysis to minimize the loss of drug that would result from the additional clearance during HD. However, performing HD immediately after dosing has been proposed

as an option for removal of toxic antibiotics^{187–189,212,217} and chemotherapeutic agents.^{218–240} For anticancer drugs, the predialysis administration of a normal dose makes sense when the patient undergoes HD 2 to 12 hours later. This strategy delivers the desired maximum plasma concentration effect while minimizing the toxic drug or metabolic effects (Table 61.7). Emerging PK and PD considerations suggest that administration after HD may not be the optimal approach for some antibacterial agents, such as aminoglycosides and vancomycin.^{187–189,212,217} High-bolus dosing immediately before or during the last hour of dialysis has been proposed for some antibiotics, but there have been few clinical studies.

If the drug is given after dialysis, the postdialysis dose (D_{HD}) should first replace the amount eliminated during the interval between dialysis sessions (D_{fail}) that is the result of clearance by the patient's residual renal function and nonrenal clearance. Furthermore, the fraction of drug removed by HD (FR) should be estimated and a supplementary dose calculated (D_{suppl}). The dose the patient should receive after HD would thus be the sum of these two doses (Fig. 61.5):

$$D_{HD} = D_{fail} + D_{suppl} = D_{fail} + (FR \times (D_{start} - D_{fail}))$$

PATIENTS RECEIVING CONTINUOUS RENAL REPLACEMENT THERAPY

CRRT and hybrid RRTs are commonly used to manage patients with AKI in intensive care units.²⁴¹ CRRT seems to

Table 61.7 Drugs Best Administered Prior to Hemodialysis

Drug Class	Examples	Drug Fraction Removed by One Dialysis Session	Reference
Anticancer	Carboplatin	20% dose post-HD; 84% dose prior to HD	Chatelut et al. ²¹⁸ , Kamata et al. ²¹⁹ , Yoshida et al. ²²⁰ , Oguri et al. ²²¹
	Cisplatin	85%	Watanabe et al. ²²²
	Oxaliplatin	65%	Katsumata et al. ²²³
	Cyclophosphamide	22% (M % unknown)	Haubitz et al. ²²⁴
	Ifosfamide	70%–87% (M, 72%–77%)	Carlson et al. ²²⁵
	Capecitabine	FBAL 50%	Walko and Lindley ²²⁶
	Gemcitabine	dFdU 50%	Koolen et al. ²²⁷
	Methotrexate	(M, 36%)	Garlich and Goldfarb ²²⁸
	Cytosine arabinoside	39% (M, 52%–63%)	Radeski et al. ²²⁹
	Topotecan	50%	Herrington et al. ²³⁰
	Bleomycin	30%–60%	Kamidono et al. ²³²
	Lenalidomide	30%	João et al. ²³³
	Pemetrexed	30%	Izzedine ²³⁴
	Tegafur = S-1	60%	Tomiyama et al. ²³⁵
	Carboplatin	84%	Oguri et al. ²³⁶
	Fludarabine	2F-Ara-A 25%	Kielstein et al. ²³⁷
	Fluorouracil = 5-FU infusion	FBAL 60%	Rengelshausen et al. ²³⁸
	Iodine 131	50%	Fofi et al. ²³⁹
	Irinotecan	SN-38 50%	Koike et al. ²⁴⁰
Aminoglycoside	Gentamicin	75%	Veinstein et al. ²¹²
	Tobramycin	80%	Kamel et al. ¹⁸⁹
Contrast agent	Gadolinium	65%–74%	Rodby ²³¹

dFdU, 2',2'-difluorodeoxyuridine; FBAL, alpha-fluoro-beta-alanine; HD, hemodialysis; M, metabolite.

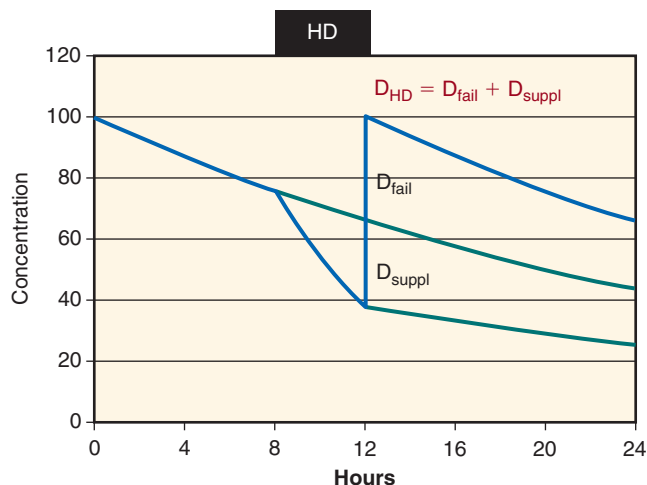


Fig. 61.5 Supplementary dose after hemodialysis (HD). To maintain therapeutic target concentrations, a supplementary dose must be given after hemodialysis to replace the removed fraction of the dose. The dose after dialysis (D_{HD}) combines both the adjusted maintenance dose (D_{fail}) and the supplementary dose (D_{suppl}).

provide less of a challenge for drug dosing than intermittent HD because its continuous nature is analogous with drug removal by native kidneys and potentially amenable to the use of standard, first-order drug clearance equations to calculate dosing. However, in practice, CRRT rarely proves as continuous as planned. The CRRT modality and details of the therapy prescription can also have significant effects on drug clearance. MW, membrane characteristics (highly variable between systems), blood flow rate, and dialysate flow rate determine the rate and extent of drug removal.²⁴² Because most drugs have MWs < 1.5 kDa, drug removal by CRRT does not depend greatly on MW. The use of higher hemofiltration volumes, especially if infused prefilter, can also affect clearance. The removal of urea, creatinine, and vancomycin was increased by 15% to 25% by the predilution modality.^{243,244}

CRRT clearances have been noted to decline because the time the hemofilter has been in use increases due to the accumulation of protein on the dialysis membrane. Clotting within the hemofilter's hollow fibers also reduces the overall surface area for clearance. Although these factors have received little direct investigation, it appears that they do affect drug clearance.^{245,246}

Drug protein binding also affects how much is removed during CRRT because only unbound drug is available for elimination by CRRT. Protein binding of >80% provides a substantial barrier to drug removal by convection or diffusion. During continuous venovenous hemofiltration, drug clearance generally approximates the ultrafiltration rate. The addition of diffusion by continuous venovenous hemodiafiltration increases drug clearance and is dependent on the ultrafiltration and dialysate flow rates. As is the case during high-flux dialysis, drug removal often parallels the removal of urea and creatinine. Thus the simplest method for estimating drug removal during CRRT is to estimate the total urea or CrCl.^{7,245,246}

Hybrid RRTs, including sustained or slow low-efficiency dialysis (SLED), EDD, continuous SLED, slow low-efficiency

daily dialysis, and slow low-efficiency daily hemodiafiltration, which use higher dialysate flow rates and shorter treatment periods (6–12 hours in duration), are frequently used as well.^{247–250} As of 2011, hybrid RRT PK data have been published for fewer than 20 drugs.⁶ The improvement of RRT machines and filters has rendered old dosing guidelines for drugs, especially antibiotics, obsolete and potentially hazardous because they are underdosed. Although there are only a few FDA or EMA official drug dosing recommendations for patients receiving CRRT, several published dosing guidelines are widely used.^{17,216,245,246} Unfortunately, these recommendations have generally not been prospectively evaluated and their influence on patient outcomes is largely unknown.

In the absence of FDA or EMA recommendations, tertiary reference sources, or any published studies relating to the handling of a drug by CRRT (common with agents that are new to the market), it may be necessary for the clinician to formulate a dosing regimen using the PK principles presented in this chapter. The lack of PK data for antibiotics during prolonged RRT and intermittent HD in critically ill patients has made the attainment and maintenance of effective concentrations a major challenge.^{250,251} If the V_D is large (>1 L/kg), there is a low likelihood that CRRT will substantially remove the drug. The use of a high-flux dialyzer or hemofilter allows for drugs with an MW < 20 kDa to be readily removed. If the clearance of the drug by CRRT or hybrid RRT is <25% of the patient's estimated total body clearance, a further dosing adjustment is probably unnecessary. By contrast, if CRRT or hybrid RRT results in an augmentation of drug clearance by 25% to 50%, an LD based on the patient's estimated volume status should be given, and maintenance doses similar to those given to a patient with a CrCl of 30 to 50 mL/min can be used. Such estimates obviously have to take into account changing volume status and be supplemented by regular drug concentration measurements, if technically feasible.

PATIENTS UNDERGOING PERITONEAL DIALYSIS

Peritoneal dialysis, as practiced in 2018, is very unlikely to enhance total body clearance of any drug by >10 mL/min because most typical peritoneal dialysis prescriptions can achieve a urea clearance of about ≤10 mL/min. As most drugs are larger than urea, their clearance is even less; thus it is very likely to be from 5 to 7.5 mL/min or less. Many studies performed in the 1970s and 1980s showed that drug clearances by peritoneal dialysis were in this very low range, so one can conclude that peritoneal dialysis does not enhance drug removal to a degree that would require a special dosage regimen modification.^{252–254} Accordingly, when given only once a day with the first dialysis bag, antibiotics are well absorbed from the peritoneum into the systemic circulation. The systemic drug concentrations will persist for the rest of the day during the three subsequent changes of peritoneal fluid. Thus oral or IV drug therapy recommendations for patients with an eCrCl or eGFR <15 mL/min are likely clinically useful.

Intraperitoneal drug administration is well accepted for the treatment of peritoneal dialysis-associated peritonitis and other infections.^{254,255} Administration intervals depend on the half-life of the drug, which is mainly determined by residual renal and extrarenal metabolic clearance. Long-standing

experience with intermittent antibiotic administration exists for the glycopeptides vancomycin and teicoplanin, which can be administered at 5- to 7-day intervals, as well as for aminoglycosides and cephalosporins, which are suitable for once-daily dosing.^{256,257}

Patients treated by automated peritoneal dialysis (APD), with frequent short-dialysis cycles, may achieve higher plasma concentrations as compared with antibiotic loading in a single extended dwell period in patients on continuous ambulatory peritoneal dialysis (CAPD). Conversely, the higher dialysate flow and small-molecule clearance achieved with APD regimens may lead to a greater peritoneal clearance of antibiotic in the intervals between dosing.²⁵⁶

Because most PK studies establishing peritoneal antibiotic doses have used 4- to 8-hour loading periods, it is recommended to perform antibiotic loading by an extended cycle in both CAPD and APD patients. For intermittent maintenance dosing, a long nighttime dwell time should be used in CAPD patients and a long daytime dwell time in APD patients. In clinical practice, intraperitoneal antibiotic dosing has not been unequivocally successful in eradicating bacterial growth, partially questioning the concept of antibiotic back diffusion into the peritoneal cavity. Thus antibiotics may need to be administered IV to enhance the antimicrobial effect.

Clinical Relevance

Hemodialysis removal of therapeutic drugs can dramatically impact the clinical outcomes of patients. High rates of removal can lead to therapeutic failures, morbidity, and even death. The advances in dialyzer membrane technology continue to increase the efficiency of the dialysis process and thus therapeutic failures become more common, especially if clinicians base their prescribing practices on data derived from low-efficiency dialyzers that are no longer clinically available.

CLINICAL BOTTOM LINE

Recommendations for dosing selected drugs in patients with CKD and AKI are given in [Table 61.8](#). These are meant only as a guide and do not imply the safety or efficacy of a recommended dose in an individual patient. An LD equivalent to the usual dose in patients with normal kidney function should be considered for drugs with half-lives >12 hours. No controlled clinical trials have established the efficacy of these dosage recommendations. The effect on drug removal of HD, ambulatory peritoneal dialysis, and CRRT is variable and the values in the table are more qualitative than quantitative. Most of these recommendations were established before high-efficiency HD treatments were practical, continuous cycling nocturnal peritoneal dialysis was common, and diffusion was added to hemofiltration in CRRT.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

Table 61.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury

Degree of Drug Dose Reduction or Interval Prolongation				Dosage Recommendations for Patients Receiving Renal Replacement Therapy		
Drug	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min	HD	CAPD	CRRT
Acetubitolol	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
	q4h	q6h	q8h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
	q6h	q12h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
	Avoid	Avoid	Avoid	Avoid	Avoid	Avoid
Acetohydroxamic acid	100%	100%	Avoid	Unknown	Unknown	Unknown
Acetylsalicylic acid	q4h	q4–6h	Avoid	As normal GFR	As normal GFR	Dose as GFR 10–50
Acrivastine	8 mg q6h	8 mg q8–12h	8 mg q12–24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Acyclovir	5 mg/kg q8h	5 mg/kg q12–24h	2.5–5 mg/kg q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Allopurinol	100%	50%	33%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Amantadine	q24h	q48–72h	q7days	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Amikacin ^a	5–6 mg/kg q12h	3–4 mg/kg q24h	2 mg/kg q24–48h	5 mg/kg after HD	15–20 mg/L/day	7.5 mg/kg q24h
Amiloride	100%	50%	Avoid	NA	NA	NA
Amlodipine	100%	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR < 10
Amoxapine	100%	100%	100%	Unknown	Unknown	Dose as GFR 10–50
Amphotericin	q24h	q24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Amphotericin B	q24h	q24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose for GFR 10–50
Amphotericin B lipid	q24h	q24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ampicillin	250 mg-2 g q4–6h	250 mg-2 g q6h	250 mg-1 g q6h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Apixaban	100%	25–50 mL/min: 100% <25 mL/min: Not recommended	Not recommended	5 mg BID; 2.5 mg BID if age >79 years or body weight <61 kg	Unknown	Unknown
Aripiprazole	100%	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR < 10
Atenolol	100% q24h	50% q24h	25% q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Auranofin	6 mg q24h	3 mg q24h	Avoid	Avoid	Avoid	Dose as GFR 10–50
Azathioprine	100%	75%–100%	50%–100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Aztreonam	100%	50%	25%	Dose as GFR < 10	Dose for GFR < 10	Dose as GFR 10–50
Benazepril	100%	50%–75%	25%–50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Bezafibrate	50%–100%	25%–50%	Avoid	200 mg q72h	200 mg q72h	200 mg q24–48h
Bisoprolol	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Bleomycin	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Bretylum	100%	25%–50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Bumetanide	100%	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR < 10
Bupropion	100% q24h	100% q24h	100% q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Butorphanol	100%	75%	50%	Unknown	Unknown	Dose as GFR 10–50
Canagliflozin	100%	30–50 mL/min: Not Recommended <30 mL/min: Avoid	Avoid	Avoid	Unknown	Unknown
Capreomycin	q24h	q24h	q48h	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Captopril	100% q8–12h	75% q12–18h	50% q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Carboplatin	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50

Carteolol	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Cefaclor	100%	100%	250–500 mg q8h	250–500 mg q8h	Dose as GFR 10–50
Cefadroxil	q12h	q12h	0.5–1.0 g after HD	0.5–1.0 g after HD	Dose as GFR 10–50
Cefamandole	q8h	q6–8h	0.5–1.0 g q12h	0.5–1.0 g q12h	Dose as GFR 10–50
Cefazolin	q8h	q12h	15–20 mg/kg after HD	15–20 mg/kg after HD	Dose as GFR 10–50
Cefepime	q12h	50%–100% q24h	Dose as GFR < 10	Dose as GFR < 10	1–2 g q12h
Cefixime	100%	75%–100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Cefotaxime	q6h	q6–12h	Dose as GFR < 10	Dose as GFR < 10	1–2 g q12h
Cefotetan	q12h	q24h	1 g after HD	1 g after HD	Dose as GFR 10–50
Cefoxitin	q6–8h	q8–12h	1 g after HD	1 g after HD	Dose as GFR 10–50
Cefpodoxime	100%	100%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Cefprozil	100%	50% q12h	250 mg after HD	250 mg after HD	Dose as GFR < 10
Ceftazidime	100%	1–2 g q24h	1 g after HD	1 g after HD	1–2 g q12h
Ceftibuten	100%	50%	400 mg after HD	400 mg after HD	Dose as GFR 10–50
Ceftizoxime	q8h	q12h	1 g after HD	1 g after HD	Dose as GFR 10–50
Cefuroxime (IV)	100% q8h	q8–12h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Celiprolol	100%	100%	50% q12h	50% q12h	Dose as GFR 10–50
Cephalexin	250–500 mg q6h	250–500 mg q8–12h	50% q12h	50% q12h	Dose as GFR 10–50
Cephadrine	100%	50%	0.5–1 g q24h	0.5–1 g q24h	Dose as GFR 10–50
Cetirizine	100%	100%	25%	25%	Dose as GFR 10–50
Chloroquine	100%	100%	750 mg q12h	750 mg q12h	As normal GFR
Chlorpropamide	50%	Avoid	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Chlorthalidone	q24h	Avoid	Avoid	Avoid	As normal GFR
Cibenzoline	100% q12h	100% q12h	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Cidofovir	50%–100%	Avoid	No data	No data	As normal GFR
Cilazapril	75% q24h	50% q24–48h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Cimetidine	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ciprofloxacin	100%	50%–100%	250 mg q12h	250 mg q12h	200 mg IV q12h
Cisplatin	100%	75%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Clarithromycin	100%	75%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Clodronate	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Clofazimine	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Clofibrate	q6–12h	q12–18h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Clomipramine	100%	Start at lower dose, monitor effect	Dose as GFR 10–50	Dose as GFR 10–50	Dose as GFR 10–50
Clonidine	q12h	q12–24h	As normal GFR	As normal GFR	As normal GFR
Clopidogrel	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Codeine	100%	75%	As normal GFR	As normal GFR	As normal GFR
Colchicine	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Cyclophosphamide	100%	75%–100%	50%	50%	100%
Cycloserine	q12h	q12–24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Dabigatran	100%	15–50 mL/min: 75 mg BID	Unknown	Unknown	Unknown
Dapson	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Daunorubicin	100%	75%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Didanosine	50%–100%	33%–50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Diflunisal	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Digitoxin	100%	100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Digoxin ^a	100% q24h	25%–50% q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Disopyramide	q8h	q12h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50

Table 61.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Cont'd)

Dosage Recommendations for Patients Receiving Renal Replacement Therapy						
Drug	Degree of Drug Dose Reduction or Interval Prolongation			HD	CAPD	CRRT
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min			
Dobutamine	100%	100%	100%	As normal GFR	As normal GFR	As normal GFR
Doxacurium	100%	50%	50%	Unknown	Unknown	Dose as GFR 10–50
Dyphylline	75%	50%	25%	Dose as GFR < 10	Unknown	Dose as GFR 10–50
Emtricitabine	q24h	q48–72h	q96h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Enalapril	100%	50%–100%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ertapenem	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Erythromycin	100%	100%	50%–75%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Ethambutol	q24h	q24–36h	q48h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ethchlorvynol	100%	Avoid	Avoid	Dose as GFR < 10	Dose as GFR < 10	NA
Ethionamide	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ethosuximide	100%	100%	75%–100%	As normal GFR	As normal GFR	As normal GFR
Etoposide	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Famciclovir	100%	q12–24h	50% q24–48h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Famotidine	100%	50%	20 mg q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Fentanyl	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Fexofenadine	q12h	q12–24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Flecainide	100%	50%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Fluconazole	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Flucytosine	50 mg/kg q12h	50 mg/kg q24h	50 mg/kg q24–48h	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Fludarabine	75%–100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Foscarnet	28 mg/kg/q8h	15 mg/kg/q8h	6 mg/kg/q8h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Fosinopril	100%	100%	75%–100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Gabapentin	300–600 mg q8h	200–700 mg q12h	100–300 mg q24h	LD: 300 mg MD: 100–300 mg q24h	As normal GFR	As normal GFR
Gallamine	75%	Avoid	Avoid	NA	NA	Avoid
Ganciclovir	2.5–5 mg/kg q12h	1.25–2.5 mg/kg q24h	1.25 mg/kg q24h	Dose as GFR < 10	Dose as GFR < 10	2.5 mg/kg q24h
Gemfibrozil	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Gentamicin ^a	5–7 mg/kg/day	2–3 mg/kg/day by levels	2 mg/kg q48–72h by levels	3 mg/kg after HD by levels	3–4 mg/L/day by levels	Dose as GFR 10–50 by levels
Gliclazide	50%–100%	20–40 mg/day	20–40 mg/day	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Glipizide	100%	50%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR < 10
Guanadrel	q12h	q12–24h	q24–48h	Unknown	Unknown	Dose as GFR 10–50
Guanethidine	q24h	q24h	q24–36h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Hydralazine	q8h	q8h	q8–12h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Hydroxyurea	100%	50%	20%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Hydroxyzine	100%	50%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Idarubicin	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ifosfamide	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Iloprost	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Imipenem	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Indapamide	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Indobufen	100%	50%	50%	Dose as GFR < 10	Dose as GFR < 10	NA
Isoniazid	100%	100%	75%–100%	Unknown	Unknown	Unknown
				Dose as GFR < 10	Dose as GFR < 10	As normal GFR

Kanamycin ^a	7.5 mg/kg q12h	7.5 mg/kg q24–72h	7.5 mg/kg q48–72h	50% the normal dose	15–20 mg/L/day	Dose as GFR 10–50
Ketorolac	100%	50%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Lamivudine	100%	50–150 mg q24h	25–50 mg q24h	Dose as GFR < 10	Dose as GFR < 10	50 mg q24h
Lenalidomide	100%	10 mg q24h for 30–60 mL/min	Avoid	5 mg q24h after HD	Avoid	Avoid
Lepirudin	100%	15 mg q48h for 10–29 mL/min				
Levofloxacin	100%	25%–50%	Avoid	Avoid	Avoid	Avoid
Lincomycin	q6h	50%	25%–50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Lisinopril	100%	q6–12h	q12–24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Lithium carbonate ^a	100%	50%–75%	25%–50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Lithium	100%	50%–75%	25%–50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Lomefloxacin	100%	50%–100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Loracarbef	q12h	q24h	q3–5days	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Melphalan	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Meperidine	100%	75%	50%	Avoid	Avoid	Avoid
Meprobamate	q6h	q9–12h	q12–18h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Meropenem	500 mg–2 g q8h	500 mg–1 g q12h	500 mg–1 g q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Metformin	100%	30–50 mL/min: 25%–50%	Avoid	Avoid	Avoid	Avoid
		10–29 mL/min: 25%				
Methadone	100%	100%	50%–75%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Methotrexate	100%	50%	Avoid	Avoid	Avoid	Dose as GFR 10–50
Methylidopa	q8h	q8–12h	q12–24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Metoclopramide	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Metocurine	75%	50%	50%	Unknown	Unknown	Dose as GFR 10–50
Mexiletine	100%	100%	50%–75%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Midazolam	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Midodrine	5–10 mg q8h	5–10 mg q8h	2.5–10 mg q8h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Milrinone	100%	100%	50%–75%	No data	No data	Dose as GFR 10–50
Mitomycin C	100%	100%	75%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Mivacurium	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Morphine	100%	50%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as < 10
Mycophenolate mofetil	100%	50%–100%	50%–100%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
N-Acetylcysteine	100%	100%	75%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Nadolol	q24h	q24–48h	q48h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Nalidixic acid	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Neostigmine	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Netilmicin ^a	4–7.5 mg/kg/day	3–7.5 mg/kg/day	2 mg/kg Q24h	2 mg/kg after each HD	IV: 2 mg/kg Q48h	Dose as GFR 10–50
Nicotinic acid	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Nitroprusside	100%	100%	Avoid	Avoid	Avoid	Dose as GFR 10–50
Nitrosoureas	100%	75%	25%–50%	Dose as GFR < 10	Dose as GFR < 10	Unknown
Nizatidine	75%–100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Norfloxacin	q12h	q12–24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Oxofloxacin	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Oxcarbazepine	100%	75%–100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pancuronium	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Paroxetine	100%	50%–75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50

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Table 61.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Cont'd)

Degree of Drug Dose Reduction or Interval Prolongation				Dosage Recommendations for Patients Receiving Renal Replacement Therapy		
Drug	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min	HD	CAPD	CRRT
p-Aminosalicylic acid	100%	50%–75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Penicillamine	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Penicillin G	100%	75%	20%–50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pentamidine	q24h	q24h	q24–36h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pentazocine	100%	75%	50%	Dose as GFR < 10	Unknown	Dose as GFR 10–50
Pentopril	100%	50%–75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pentoxifylline	q8–12h	q12–24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Perindopril	2 mg q24h	2 mg q24–48h	2 mg q48h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Phenobarbital	q8–12h	q8–12h	q12–16h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Phenylbutazone	100%	50%	Avoid	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pipecuronium	100%	50%	25%	Avoid	Avoid	Dose as GFR 10–50
Piperacillin	q6h	q6–12h	q12h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Plicamycin	100%	75%	50%	Unknown	Dose as GFR < 10	Dose as GFR 10–50
Pomalidomide	100%	Avoid	Avoid	Unknown	Unknown	Unknown
Pravastatin	100%	30–50 mL/min: 100% 10–30 mL/min: 10 mg q24h	10 mg q24h	Dose as GFR < 10	Unknown	Unknown
Pregabalin	100% q8–12h	50% q8–12h	25% q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Primidone	q12	q12–24h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Probenecid	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Procainamide	q4h	q6–12h	q8–24h	Follow levels	Dose as GFR < 10	Dose as GFR 10–50
Propoxyphene	100%	100%	Avoid	Avoid	Avoid	Avoid
Propylthiouracil	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pyrazinamide	100%	100%	50%–100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Pyridostigmine	100%	35%	20%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Quinapril	100%	2.5–5 mg q24h	2.5 mg q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Quinine	q8h	q8–12h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ramipril	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ranitidine	100%	150 mg q24h	75 mg q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Ribavirin	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Rifampin	100%	50%–100%	50%–100%	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Rivaroxaban	100%	15–50 mL/min 15 mg q24h	Avoid	Avoid	Avoid	As normal GFR
Simvastatin	100%	100%	5 mg q24h	Dose as GFR < 10	Dose as GFR < 10	As normal GFR
Sitagliptin	100%	30–50 mL/min: 50% 10–30 mL/min: 25%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Sotalol	100%	25%–50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Spironolactone	100%	Usual dose q12–24h	Avoid	Avoid	Dose as GFR < 10	Dose as GFR 10–50
Stavudine	100%	50% q12–24h	50% q24h	Avoid	Avoid	Avoid
Streptomycin ^a	q24h	q24–72h	q72–96h	Dose as GFR < 10	20–40 mg/L/day	Dose as GFR 10–50
Streptozocin	100%	75%	50%	Unknown	Unknown	Unknown
Sulfamethoxazole	q12h	q18h	q24h	1 g after dialysis	1 g/day	Dose as GFR 10–50
Sulfipyrazone	100%	100%	Avoid	Avoid	Avoid	Dose as GFR 10–50
Sulfisoxazole	q6h	q8–12h	q12–24h	2 g after dialysis	3 g/day	NA
Sulindac	100%	50%–100%	50%–100%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR < 10

Sulotroban	50%	30%	10%	Unknown	Unknown	Unknown
Tazobactam	100%	75%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Teicoplanin	q24h	q24–48h	q48–72h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Temocillin	q12–24h	q24h	q48h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Terbutaline	100%	50%	Avoid	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Tetracycline	100%	100%	50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Thiazides	100%	100%	Avoid	Dose as GFR < 10	Dose as GFR < 10	NA
Thiopental	100%	100%	75%	NA	NA	NA
Ticarcillin	50–75 mg/kg q6h	50–75 mg/kg q8h	50–75 mg/kg q12h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Tobramycin ^a	5–7 mg/kg/day	2–3 mg/kg/day	2 mg/kg q48–72h	3 mg/kg after HD	3–4 mg/L/day	Dose as GFR 10–50
Tolvaptan	100%	100%	Avoid	Avoid	Avoid	Avoid
Topiramate	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Topotecan	75%	50%	25%	Dose as GFR < 10	No data	No data
Tramadol	100%	50–100 mg q8h	50 mg q8h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Tranexamic acid	50%	25%	10%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Trazodone	100%	100%	Avoid/50%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Triamterene	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Trimethoprim	q12h	q12h	q24h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Trimetrexate	100%	50%–100%	Avoid	No data	No data	Dose as GFR 10–50
Tubocurarine	75%	50%	Avoid	Unknown	Unknown	Dose as GFR 10–50
Valganciclovir	50%–100%	450 mg q24–48h	450 mg q72–96h	Avoid	Avoid	450 mg q48h
Vancomycin ^a	1 g q12–24h	1 g q24–96h	1 g q4–7days	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Venlafaxine	100%	25%–50%	25%–50%	Dose as GFR 10–50	Dose as GFR < 10	Dose as GFR 10–50
Vigabatrin	100%	50%	25%	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Zalcitabine	100%	q12h	q24h	Dose as GFR < 10	No data	Dose as GFR 10–50
Zidovudine (AZT)	100% q8h	100% q8h	50% q8h	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR 10–50
Zileuton	100%	100%	100%	Dose as GFR < 10	Unknown	Dose as GFR 10–50
Zopiclone	100%	3.75–5 mg daily	3.75–5 mg daily	Dose as GFR < 10	Dose as GFR < 10	Dose as GFR < 10

^aAdjust dose to achieve desired serum concentrations using measured serum concentrations and pharmacokinetic modeling principles.

AZT, Azidothymidine; BID, two times a day; CAPD, continuous ambulatory peritoneal dialysis; CRRT, continuous renal replacement therapy; GFR, glomerular filtration rate; HD, hemodialysis; IV, intravenous; LD, loading dose; MD, maintenance dose; NA, not applicable; qndays, every *n* days; qnh, every *n* hours.

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BOARD REVIEW QUESTIONS

- All of the following are potential mechanisms by which the volume of distribution of drugs is increased in patients with chronic kidney disease *except*
 - Decreased plasma protein binding
 - Increased tissue binding
 - Increased fluid excretion
 - Increased fluid status

Answer: c

Rationale: The absorption, distribution, metabolism, and renal excretion of many drugs is altered by chronic kidney disease. The increase in volume of distribution (V_D) may be the result of pathophysiologic alterations in body composition, fluid overload secondary to excessive fluid administration or intake, decreased protein binding, or increased tissue binding.

- Renal clearance of a drug is a composite of all of the following *except*
 - Absorption
 - Glomerular filtration rate
 - Reabsorption
 - Secretion

Answer: a

Rationale: Kidney function is the most quantifiable determinant of renal drug clearance. It is important to note that the term “kidney function” includes the combined processes of glomerular filtration, tubular secretion, and reabsorption, as well as endocrine and metabolic functions. Alterations in any or all of these functions secondary to chronic kidney disease may have a dramatic effect on drug clearance.

- Which of the following statements regarding the effects of dose and dosing interval adjustments in patients with reduced renal function is true?
 - If the dose is reduced while the dosing interval remains unchanged, the peak will be higher and the trough lower.
 - If the dose remains unchanged while the dosing interval is increased, the peak and trough concentrations in patients with reduced renal function will be similar to those in patients with normal renal function.
 - If the dose is reduced while the dosing interval remains unchanged, the peak and trough concentrations in patients with reduced renal function will be similar to those in patients with normal renal function.
 - If the dose remains unchanged while the dosing interval is increased, the peak will be higher and the trough lower.

Answer: b

Rationale: The initial dose for chronic kidney disease (CKD) patients should be the same as the dose recommended for those with normal renal function unless the drug's volume of distribution is known to be altered in the presence of CKD or a concomitant disease, in which case the dose should be increased proportionally. This is the case in order to achieve the same peak. In order for the similar trough concentrations to be achieved, the interval should be extended as the concentration–time profile of drugs is altered in patients with CKD.

- A 75-year-old woman with known atrial fibrillation is admitted to the hospital because of epistaxis while on warfarin. She needs a nasal tamponade and, because of subsequent fever, administration of cefazolin intravenously. In addition, the warfarin will be replaced by rivaroxaban. Because the serum creatinine is 200 $\mu\text{mol/L}$ (2.26 mg/dL) and her body weight is 80 kg, her estimated creatinine clearance by Cockcroft–Gault equation is 30 mL/min . Which of the following decisions is wrong?
 - The cefazolin therapy was started with 2000 mg.
 - The cefazolin maintenance dose was reduced to 30% of usual standard dose and the administration interval extended from 8 to 12 hours.
 - Rivaroxaban was started 3 days later with 20 mg/day .
 - The rivaroxaban anti-Xa activity was measured 5 days after anticoagulation.
 - Because the anti-Xa activity was > 200 ng/mL , the rivaroxaban dose was reduced to 15 mg/day .

Answer: b

Rationale: Only the cefazolin maintenance dose should be reduced, but the dosing interval should not be extended. Normal cefazolin half-life is 2.2 hours but with kidney failure it is 33 hours. Accordingly, the fraction eliminated by the kidneys, FE, is $1 - 2.2/33 = 0.93$. The dose adjustment factor, Q, can be calculated from the equation

$$Q = 1 - (\text{FE}[1 - \text{KF}]),$$

where KF is the fractional kidney function for this patient, which is 0.25 (CrCl of 30 mL/min divided by normal CrCl of 120 mL/min). From this, the dose fraction, Q, is determined to be 0.3. Because cefazolin action is considered to be time dependent, the administration interval should not be extended in addition to the dose reduction. Otherwise, the trough levels could fall below the anti-infective threshold concentration.

- Which of the following drugs is most likely to be removed by high-flux hemodialysis?
 - Amlodipine [molecular weight (MW) = 567; volume of distribution (V_D) = 21.0 L/kg ; plasma protein binding > 95%]
 - Atenolol (MW = 266; V_D = 1.1 L/kg ; plasma protein binding = 3%)
 - Clindamycin (MW = 476; V_D = 0.8 L/kg ; plasma protein binding = 94%)
 - Apixaban (MW = 460; V_D = 0.3 L/kg ; plasma protein binding = 87%)

Answer: b

Rationale: The impact of hemodialysis on a patient's drug therapy is dependent on several factors, including the physicochemical characteristics of the drug, the dialysis conditions, and the clinical situation for which dialysis is performed. Drug-related factors that affect dialyzability include the molecular weight or size, degree of protein binding, and V_D . Dialysis membranes in the 21st century are predominantly composed of semisynthetic or synthetic materials (e.g., polysulfone, polymethylmethacrylate, or polyacrylonitrile). These high-flux dialysis membranes have larger pore sizes and

more closely mimic the filtration characteristics of the human kidney. This allows the passage of most solutes, including drugs that have a molecular weight of $\leq 20,000$ Da. Drugs that are small but highly protein bound (i.e., $>90\%$) are not well dialyzed because both the principal binding proteins,

α_1 -acid glycoprotein and albumin, have a very high MW. For example, the MW of albumin is 60,000 Da. Finally, those drugs that are widely distributed with $V_D > 2$ L/kg are poorly removed by hemodialysis.